Semantic Causal Propagation: An Individuation-Theoretic Calculus of Boundary-Free Search, and a Domain-Specific Language for Accountable Research Projects

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Abstract

We develop, entirely within the language of finite weighted graphs, a formal theory of *search* as the individuation of a claim from an unbounded, non-completable medium, and a domain-specific orchestration language, *Graffiti* (`.grf`), whose executable semantics is exactly this theory. The central observation is that a search query and the act of naming or recognising the thing sought are the same operation under two readings: both are the individuation of a region of claim-space by negation against a medium that is never fully specified. From this we derive, as theorems rather than assumptions: a strictly positive *resolution floor* on every search, forcing every returned claim to be a bounded region rather than an exact point; a proof that natural-language paraphrases of one underlying claim are *representations* of a single individuation and cost no additional search effort to substitute for one another; a *path-opacity* theorem guaranteeing that the interior trajectory of a multi-step search is unconstrained provided the trajectory converges on its target, so that intermediate results may look arbitrarily unlike the sought claim without invalidating the search; a multiplicative law for the combination of independent search steps (*catalysts*), together with a coherence theorem showing that no fewer than three mutually reinforcing, independently sourced catalysts can ground a claim against revision by a single dissenting source; and a *closure* criterion for knowing when a search is finished that is strictly stronger than crossing a confidence threshold: a search is complete when every further catalyst available to it resolves to a claim-region already reached, so that no further inquiry can produce a materially different outcome. We then define Graffiti as a small orchestration language whose sole computational primitive, `seek`, is this individuation operation exposed to a script author: a mandatory exclusion clause, a target claim-region, an optional explicit catalyst chain, and a closure-based termination condition. We give the complete lexical grammar, context-free grammar, type system, and small-step operational semantics of Graffiti, prove progress and preservation for its type system, prove that its scheduler never revisits an already-closed claim-region and never starves a convergent catalyst chain, and specify its execution model as a directed acyclic graph of committed `seek` events over a heterogeneous catalyst registry that may include local search, remote retrieval, and machine-learned inference as computationally uniform operations. We validate the framework on a suite of constructed instances and prove a master equivalence tying the language's every guarantee to the single fact that the medium of search is never completable.

Keywords: search theory; individuation by negation; resolution floor; contact graphs; path opacity; representation mobility; catalytic composition; coherence; domain-specific languages; operational semantics; heterogeneous orchestration; accountable computation.

Contents

Part I
Foundations

1 Introduction

1.1 The question

What is a search? The question sounds settled by long practice — one supplies a query, a system returns candidates, and the practice of information retrieval has an extensive engineering literature on ranking, relevance, and recall (Manning et al., 2008; Salton and McGill, 1983). But the engineering literature is silent on a structural question that becomes acute the moment a search is not casual: when a user is not asking for the nearest plausible page but is constructing a research artefact — a project, auditable and reproducible, whose steps matter as much as its answer — what, precisely, does the system return, when is it entitled to stop, and what does it mean for two differently worded results to be *the same* finding?

We answer these questions from a single structural premise, argued rather than assumed: a search is conducted against a *medium* — the totality of what could bear on the query, without regard to whether it resides on the querying machine or elsewhere — and this medium is never exhausted by any finite search process. From this one fact, formalised below as non-completability (Theorem 3.1), an entire calculus follows by theorem: that no search can return an exact point rather than a bounded region; that paraphrase costs nothing because it is not new search but a change of coordinates on an already-found region; that a search's interior steps may look arbitrarily unlike its answer without invalidating it, provided the whole trajectory converges; that combining independent lines of inquiry is a multiplicative operation with a precise law of diminishing returns; and that a foundation-worthy claim requires no fewer than three independently sourced, mutually reinforcing findings, never a single authority nor even two.

1.2 The recognition/search identity

The premise that unifies the calculus is that *recognising* a thing and *searching* for it are the same operation read in two directions. To recognise an object is to place a percept into a region of meaning by ruling out everything the percept is not; to search for an object is to start from a description of what is wanted and to individuate, from the medium, a region satisfying that description. Both operations terminate at the same kind of object — a bounded region of claim-space, never a single point — and both are subject to the same lower bound on how precisely they can resolve their target. A query is not a degraded, lossy version of a perfect specification the searcher secretly possesses; it is itself an individuation, exactly as exact as recognition ever is. This identity, established rigorously in Section 4, is what licenses treating "finding Peter," "finding the founding date of an institution," and "recognising that the object in front of you is a cat" as instances of one mathematical operation.

1.3 Contributions

1. **The search medium and the resolution floor** (§2–3). We formalise the medium of search as a non-completable expanding family of contact graphs and prove, from non-completability alone, that every search carries a strictly positive floor: no search returns a point, only a region.
2. **Individuation by negation and the identity of recognition and search** (§4). We prove that in a medium with no privileged claim, a claim can be fixed only as the complement of what it is not, and that this is the same operation whether read as recognition (decoding a percept) or as search (projecting a description).
3. **Representation mobility** (§5). We prove that paraphrases of a fixed claim are equivalent representations under an averaging constraint, that switching between them commits no new search cost, and that no representation is privileged over another.
4. **Path opacity and convergence admissibility** (§6). We prove that a multi-step search trajectory is admissible if and only if it converges on its target, with interior steps entirely unconstrained; two trajectories sharing endpoints are indistinguishable by any test of the output.
5. **Catalytic composition and the coherence theorem** (§7). We derive the multiplicative law for combining independent search steps, a Borel–Cantelli-type saturation dichotomy, and prove that a claim is robustly grounded if and only if its supporting catalysts form a strongly connected structure of at least three members.
6. **The closure criterion** (§8). We define *closure* of a search — the state in which no further catalyst resolves to a materially different claim-region — and prove it is strictly stronger than, and the correct replacement for, a simple confidence threshold.
7. **The Graffiti language** (§9–§11). We give the complete lexical grammar, context-free grammar, static type system, and small-step operational semantics of a language whose sole primitive, `seek`, is the individuation operation of §4–§8 exposed to a script author, and

prove type soundness and scheduler soundness.

8. **The orchestration model** (§12). We specify Graffiti's execution model as a directed acyclic graph of committed `seek` events dispatched to a heterogeneous catalyst registry — local computation, remote retrieval, and machine-learned inference treated as uniform operations — and prove the scheduler is livelock-free and never re-executes a closed claim.

1.4 Method

Every object in this paper is a finite weighted graph or a construction on one, exactly as in the classical theory of network flows and cuts (Ahuja et al., 1993; Ford and Fulkerson, 1956; Menger, 1927). We introduce no metric space, no measure, and no probability calculus beyond a derived normalisation on $[0, 1]$. This is a discipline, not an aesthetic: every claim reduces to a statement about vertices, edges, and weights, and is therefore checkable by direct computation — minimum cuts are exactly computable by maximum flow (Ford and Fulkerson, 1956; Goldberg and Tarjan, 1988). Where we use a word with an evocative reading — "claim," "search," "catalyst" — the word names a specific graph-theoretic object defined in the text, and no theorem depends on any reading beyond that definition. Interpretive remarks are marked as such and are never load-bearing.

1.5 Organisation

Section 2 fixes the contact graph and the medium. Section 3 derives the resolution floor from noncompletability. Section 4 proves individuation by negation and the recognition/search identity. Section 5 proves representation mobility. Section 6 proves path opacity and convergence admissibility. Section 7 develops the catalytic algebra and the coherence theorem. Section 8 defines and characterises closure. Section 9 gives the Graffiti lexical and context-free grammar. Section 10 gives the static type system and proves soundness. Section 11 gives the operational semantics and proves determinism, monotonicity, and scheduler soundness. Section 12 specifies the heterogeneous execution model. Section 13 gives worked programs. Section 14 reports a numerical validation suite. Section 15 states scope and concludes.

2 The Search Medium

2.1 Contact graphs

Definition 2.1 (Finite weighted graph)**.** A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ where V is a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ is a set of undirected edges, and $w\colon E \to \mathbb{R}_{>0}$ is a strictly positive edge weight. We assume Γ connected unless stated. Graphs are simple (no loops or multi-edges).

Definition 2.2 (Contact graph; medium vertex)**.** A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$ together with a distinguished vertex $\mathsf{m} \in V$, the *medium*, adjacent to every other vertex: $\{v, \mathsf{m}\} \in E$ for all $v \neq \mathsf{m}$. Vertices $v \neq \mathsf{m}$ are *claims*: candidate individuated units of information — a fact, a document, an entity, a numeric value, or any other discrete thing a search may return. An edge $\{u, v\}$ between two claims is a *contact*, and its weight $w(\{u, v\})$ is the cost of the distinction separating u from v. The medium vertex m stands for the undifferentiated totality of everything not yet individuated — every source, local or remote, that has not yet been resolved into a claim.

Remark 2.3 (Why a single medium vertex, and why no boundary between local and remote). The definition draws no distinction between a claim recoverable from a file on the querying machine and a claim recoverable from a remote server: both are, prior to being individuated, contents of the single vertex m. This is a deliberate structural commitment, not an approximation: the theory that follows attaches no special status to physical locality, because locality is not what makes a claim distinguishable from the rest of the medium — the cost of individuation (Theorem 2.4) is. Two facts equally costly to establish are equally individuated regardless of where the bytes that established them are stored. This licenses the design principle of Section 12: a script's catalysts may draw on local files, remote APIs, or machine-learned models as computationally uniform sources, because the graph never encodes a location, only a cost.

2.2 Separation cost

Definition 2.4 (Separation cost)**.** The *separation cost* of a claim v is the minimum weight of a cut placing v on one side and the medium on the other,

$$\sigma(v) \;=\; \min_{\substack{S \ni v \\ \mathsf{m} \notin S}} w(\partial(S)), \qquad \partial(S) \;:=\; \{\, e = \{u, w\} \in E : u \in S,\ w \notin S \,\}.$$

Because V is finite, the minimum is attained; the minimising side $S^*(v)$ and the minimising cut $\partial(S^*(v))$ constitute the *minimum cut* of v against the medium, computable exactly by maximum flow (Ford and Fulkerson, 1956; Menger, 1927).

The separation cost is the graph-theoretic content of "how hard it is to tell v apart from everything else." A claim recoverable from a single uncontested primary source has low separation cost; a claim that can only be established by triangulating many weakly corroborating traces has high separation cost. Nothing in Theorem 2.4 presumes what kind of process realises a contact edge — Section 7 shows the edges themselves are supplied by *catalysts*, and Section 12 shows a catalyst may be a search engine query, a local file scan, or an inference performed by a trained model.

3 The Resolution Floor

We now derive the central constant of the theory. It is not posited: it follows from the conjunction of two facts already implicit in Section 2 — a claim is a proper part of the medium, and the medium is never exhausted by any finite stage of search.

3.1 Non-completability

Definition 3.1 (Expanding search family; non–completable medium)**.** An *expanding search family* is a sequence of contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \cdots$, each obtained from the previous by adding claims and contacts of weight $\geq \beta$ (fixed below) while retaining the medium vertex and all prior claims and contacts; Γ_k represents the state of search after k committed search steps. The medium is *non-completable* if the family has no terminal stage: for every finite Γ_k there exists $\Gamma_{k+1} \supsetneq \Gamma_k$. This is an order condition — no last stage of search — and not a cardinality assumption about how many claims exist.

Remark 3.2 (Why search is non-completable)*.* A search medium spanning both a local machine and the unbounded remote network admits, for any finite state of search, a further source not yet consulted: another document, another database, another model that could be queried, another cross-reference not yet followed. Theorem 3.1 records only this order fact and nothing stronger; we make no claim about the cardinality of the medium, only that no finite search process exhausts it. This is the formal content of "there is no boundary between what is on the machine and what is on the internet": the absence of a boundary is exactly the absence of a terminal stage.

3.2 Identification and its residual

Definition 3.3 (Identification; identification residual)**.** To *identify* a claim v at search stage Γ_k is to determine it by its minimum cut against the medium, $\partial(S^*(v))$, computed within Γ_k. The identification is *complete* if every claim of the whole medium has entered the computation of that cut — equivalently, if Γ_k already contains the whole medium. The *identification residual* $\rho(v) \geq 0$ is the cut content a complete identification would still remove that the finite-stage identification in Γ_k cannot; $\rho(v) = 0$ if and only if the identification is complete.

3.3 The floor theorem

Theorem 3.4 (Floor from non-completability)**.** *Let the medium be non-completable and let v be a claim ($v \neq \mathsf{m}$, and v does not exhaust V). Then the identification of v cannot be completed at any finite search stage, and $\rho(v) > 0$.*

Proof. Suppose $\rho(v) = 0$, so that identification of v is complete at some finite stage Γ_k. By Theorem 3.3, completeness means every claim of the medium has been brought to bear in the cut of v against m, i.e. the cut is computed in a graph containing the whole medium; hence Γ_k contains the whole medium. This contradicts non-completability (Theorem 3.1), which guarantees a further stage $\Gamma_{k+1} \supsetneq \Gamma_k$ containing a claim not present in Γ_k and therefore not yet brought to bear on the cut of v. Consequently no finite stage completes the identification of v, and $\rho(v) > 0$. □

Theorem 3.5 (Resolution floor)**.** *There is a constant $\beta > 0$ such that $w(e) \geq \beta$ for every contact $e \in E$ and $\sigma(v) \geq \beta$ for every claim v: no two claims are separated at zero cost, and no claim is individuated for free.*

Proof. Set $\beta := \inf_v \rho(v)$. By Theorem 3.4, $\rho(v) > 0$ for every claim v; the residual is the irreducible boundary content of v's identification and is carried by the edges of v's minimum cut, so each such edge has weight at least the per-claim residual and hence at least β. Because every contact edge $\{u, v\}$ is a boundary edge of the identification of u (it lies in $\partial(\{u\})$), every edge weight satisfies $w(e) \geq \beta$. Connectivity of Γ forces $\partial(S^*(v)) \neq \varnothing$ for every claim v, so $\sigma(v) = w(\partial(S^*(v))) \geq \beta > 0$. □

Corollary 3.6 (No sharp result)**.** *No search returns a claim separated from the medium by a cut of weight zero. Every returned claim deposits boundary content $\geq \beta$; a search that appears to answer "exactly" has, in fact, resolved to a region whose width is bounded below by β and no less.*

Remark 3.7 (The floor is a fact about the medium, not a defect of the search process). Theorem 3.5 does not say search is imprecise because search engines are imperfect; it says a search over a genuinely inexhaustible medium cannot, even with an ideal instrument, terminate at a claim of zero residual. A "perfect" search — one returning a point rather than a region — would require completing a comparison against a medium that, by hypothesis, is never completed. Section 8 makes this the basis of an honest stopping rule rather than a caveat to be apologised for.

Part II
The Search Calculus

4 Individuation by Negation and the Recognition/Search Identity

4.1 No privileged claim

Definition 4.1 (Selector; no privileged claim)**.** A *selector* for a claim set $U \subseteq V$ is a rule that fixes U by naming a distinguished member of U in advance of any search. A determination of U is *selector-free* if it names no member of U. The medium has *no privileged claim* if the data available to a determining rule are invariant under the weighted automorphism group $\mathrm{Aut}(\Gamma)$, so that no claim is singled out by the data alone.

A medium with no privileged claim is the formal statement of a search domain in which nothing is pre-marked as the answer: a claim must be picked out by the structure of the query and the medium together, not by an oracle that already knows which vertex is sought.

4.2 Individuation by negation

Theorem 4.2 (Individuation by negation)**.** *In a medium with no privileged claim, any selector-free determination of a claim set U fixes U as the complement of its negation set, $U = V \setminus \complement U$, with $\complement U := V \setminus U$ specified first; and conversely every admissible negation set $\complement U$ determines a unique U. No selector-based rule fixes U without itself invoking a further selector.*

Proof. Necessity. To fix U is to decide, for every claim of V, whether it belongs to U or to the remainder. A *positive* rule — one asserting membership of a distinguished claim of U — must name that claim; since the available data are $\mathrm{Aut}(\Gamma)$-invariant (Theorem 4.1), no claim is named by the data alone, so the rule must itself supply a selector. That selector must in turn be fixed, either given in advance (excluded by the no-privileged-claim hypothesis) or supplied by a prior rule facing the identical dichotomy. The regress terminates only at a rule that names no member of U: specifying the negation set $\complement U = V \setminus U$ and taking U as what remains. This names only "the rest," a datum already supplied once the medium V is given.

Sufficiency and uniqueness. Complementation $U \mapsto V \setminus U$ is an involution on subsets of the finite set V: $V \setminus (V \setminus U) = U$. Hence a given negation set $\complement U \subsetneq V$ with $V \setminus \complement U \neq \varnothing$ determines the unique claim set $U := V \setminus \complement U$, and conversely. □

Corollary 4.3 (A claim is fixed by its non-neighbours)**.** *A single claim v is determined selector-free by the partition of the rest of the medium into its neighbours $N(v)$ (the claims it contacts) and its non-neighbours $V \setminus N[v]$ (the claims it neither is nor contacts). No intrinsic label distinguishes v; its identity is carried by this negation structure.*

Proof. Apply Theorem 4.2 with $U = \{v\}$, refined by adjacency: the selector-free datum fixing $\{v\}$ is $V \setminus \{v\}$ partitioned into $N(v)$ and $V \setminus N[v]$; no member of $\{v\}$ itself is named by this datum. □

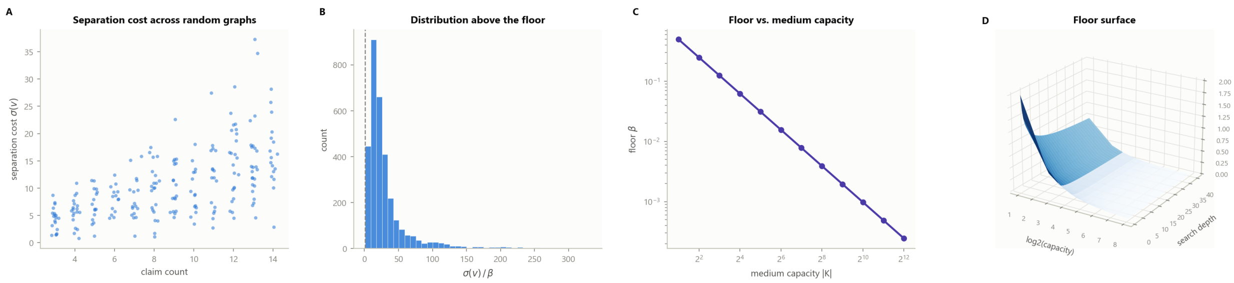

Figure 1: **The Resolution Floor.** (**A**) Separation cost $\sigma(v)$ against claim count, one point per claim across 220 random contact graphs of 3–14 claims (steelblue). Cost rises with claim count but never falls below the graph's own floor β, the empirical content of Theorem 3.5: no claim is individuated for free, regardless of how large the medium is. (**B**) Distribution of the ratio $\sigma(v)/\beta$ across 400 random graphs and all their claims; the dashed line marks $\sigma/\beta = 1$. The histogram is supported entirely at and above the floor: no claim of any constructed instance is separated from the medium below the theorem's bound (Corollary 3.6). (**C**) Floor value against medium capacity $|K|$ on log-log axes, for twelve capacities $|K| \in \{2, 4, \ldots, 4096\}$. The floor falls monotonically as capacity grows but remains strictly positive at every tested capacity, confirming that a larger medium narrows, but never closes, the resolution floor. (**D**) Three-dimensional surface of the floor over medium capacity (log scale) and search depth: the surface is bounded below by the inverse-capacity term and decays toward it as depth increases, never reaching zero. The floor is a property of the medium's finiteness, not of how long the search runs.

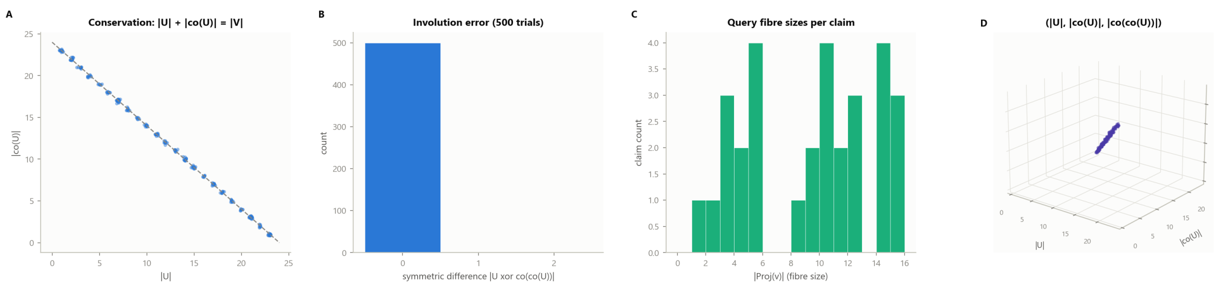

Figure 2: **Individuation by Negation.** (**A**) Subset size $|U|$ against complement size $|\complement U|$ for 260 random subsets of a 24-claim medium (steelblue); the dashed line is the conservation identity $|U| + |\complement U| = |V|$. Every point lies exactly on the line: a claim set and its negation set partition the whole medium exactly, with no residual and no overlap (Theorem 4.2). (**B**) Distribution of the symmetric-difference error $|U \text{ xor } \complement\complement U|$ over 500 trials: a single spike at 0, confirming complementation is an exact involution — double negation recovers the original claim set with no error. (**C**) Fibre size distribution: the number of distinct queries $|\Xi(v)|$ resolving to each of 30 synthetic claims (seagreen), after randomly augmenting each claim's query set. Fibre sizes vary freely across claims, the operational content of a decoder's many-to-one query structure. (**D**) Three-dimensional scatter of $(|U|, |\complement U|, |\complement\complement U|)$ across 180 random subsets (violet): every point lies on the identity plane $|\complement\complement U| = |U|$, the geometric realisation of the involution proved algebraically in panel B.

4.3 Decoding and projection: two readings of one map

We now make precise the identity between recognising a claim (decoding a given percept or query into its cell) and searching for a claim (projecting a description onto the region of the medium that satisfies it). We show these are literally the same function under an inverse reading, not merely analogous procedures.

Definition 4.4 (Query; decoder; projector)**.** Fix a contact graph Γ. A *query* is any finite string, structured value, or percept q presented to the search process. A *decoder* is a map $\mathrm{Dec}\colon Q \to V$ from the space Q of admissible queries to claims of Γ, sending a query to the claim it individuates by Theorem 4.3. A *projector* is a map $\Xi\colon V \to 2^Q \setminus \{\varnothing\}$ sending a claim to the non-empty set of queries that decode to it: $\Xi(v) := \mathrm{Dec}^{-1}(\{v\})$.

Theorem 4.5 (Recognition/search identity)**.** *For every claim $v \in V$ and every query $q \in Q$:*

$$\mathrm{Dec}(q) = v \quad \Longleftrightarrow \quad q \in \Xi(v).$$

Consequently recognition *— computing* $\mathrm{Dec}(q)$ *for a given percept q — and* search *— computing a representative element of $\Xi(v)$ for a given target claim v, or equivalently testing membership $q \in \Xi(v)$ for a candidate query q — are inverse readings of the single relation $\{(q, v) \in Q \times V : \mathrm{Dec}(q) = v\}$. Neither is definable without the other: Ξ is literally the fibre structure of* Dec, *and* Dec *is the (well-defined, by Theorem 4.3) function whose fibres are the sets $\Xi(v)$.*

Proof. By Theorem 4.4, $\Xi(v) = \mathrm{Dec}^{-1}(\{v\})$; membership $q \in \Xi(v)$ is by definition of preimage equivalent to $\mathrm{Dec}(q) = v$. This equivalence is definitional, not a coincidence to be separately established, so the two directions state the same fact. The claim that recognition and search are inverse readings of one relation follows because a decoder is exactly a function and a projector is exactly its fibre map; a function and its fibre map determine each other on a finite domain: Dec is recovered from $\{\Xi(v)\}_{v\in V}$ as the unique map sending each q to the v with $q \in \Xi(v)$ (well defined since the $\Xi(v)$ partition Q under a total decoder), and Ξ is recovered from Dec by Theorem 4.4. □

Remark 4.6 (What this identity licenses)*.* Theorem 4.5 is the formal content of the observation that motivates this paper: a thought about a fixed object is the same thought whether triggered by perceiving the object (recognition, computing Dec forward) or by being asked to find it (search, computing a representative of Ξ backward). A search engine and a classifier are, on this reading, the same computational object queried in opposite directions. This licenses a single language primitive (Section 9) for both operations: a `seek` expression names a target region and asks the runtime to exhibit a query (a chain of catalyst invocations) whose decode lands there, which is exactly computing an element of Ξ for that region.

5 Representation Mobility

We now address paraphrase directly: many syntactically distinct descriptions can individuate the same claim, and switching among them must not be confused with performing additional search.

5.1 Representations of a claim

Definition 5.1 (Representation; representation fibre)**.** Fix a claim v and a target claim x^* against which alignment is measured (Theorem 6.1 below formalises alignment; for now take $a(v, x^*) \in (0, 1]$ as a fixed real). A *representation* of v at dimension N is a tuple $\mathsf{R}(v) = (s_1, \ldots, s_N) \in \mathbb{R}^N$ satisfying the *averaging constraint*

$$\frac{1}{N}\sum_{j=1}^{N} s_j \;=\; a(v, x^*),$$

with individual components s_j otherwise unconstrained in $\mathbb{R}$ (in particular, a component may lie outside $(0, 1]$). The *representation fibre* of v at dimension N is the set of all such tuples.

Theorem 5.2 (Representation mobility)**.** *For every claim v and every $N \geq 2$:*

(i) *the representation fibre of v at dimension N is non-empty and infinite;*

(ii) *switching from one representation of v to another commits no new contact edge and leaves the committed search-step count unchanged (Theorem 11.1 below);*

(iii) *any downstream computation that depends on v only through the average $a(v, x^*)$ is invariant under representation switching.*

Proof. (i) The averaging constraint is a single linear equation in N unknowns; for $N \geq 2$ its solution set is an affine hyperplane of $\mathbb{R}^N$, hence non-empty and infinite, and any $(s_1, \ldots, s_{N-1}) \in \mathbb{R}^{N-1}$ extends uniquely to a solution by $s_N = N\, a(v, x^*) - \sum_{j<N} s_j$.

(ii) A representation switch replaces one tuple in v's fibre with another; both denote the same claim v at the same alignment $a(v, x^*)$ (both satisfy the same averaging constraint), so the switch introduces no new claim, no new contact, and hence no new committed search step (Theorem 11.1).

(iii) By hypothesis the downstream computation is a function of $a(v, x^*)$ alone, and every member of v's representation fibre averages to the same value, so the computation's result is unchanged. □

Example 5.3 (The stash example)**.** Let v be a fixed individual, and let x^* be the claim "is identifiable as this individual." The strings "Peter," "a Harvard medical student," and "a specialist in the anatomical processes of diseased human physiology, training at an institution of higher learning in Boston" are three representations of v in the sense of Theorem 5.1: each, decoded by an appropriately informed reader, individuates the same claim v against the same target x^*, and each satisfies the same averaging constraint even though their surface content — their component-wise detail — differs enormously. By Theorem 5.2(ii), producing the third string having already established the first commits no new search: it is a change of representation on an already-individuated claim, not a verification of a new one.

Theorem 5.4 (Receiver-relative decoding is not error)**.** *Let* $\mathrm{Dec}_1, \mathrm{Dec}_2$ *be two decoders (Theorem 4.4) over graphs* Γ_1, Γ_2 *with distinct edge sets, modelling two readers with different background knowledge. There exist a query* q *(e.g. a representation of* v *as in Theorem 5.3) and claims* $v_1 \neq v_2$ *such that* $\mathrm{Dec}_1(q) = v_1$ *and* $\mathrm{Dec}_2(q) = v_2$*, with both decodings individuating a claim of separation cost* $\geq \beta$ *within their respective graphs; neither decoding is in error.*

Proof. Because $\Gamma_1 \neq \Gamma_2$ as weighted graphs, the same query q may individuate different claims under Dec_1 and Dec_2 (the two decoders act on structurally different negation graphs, and Theorem 4.2 determines U relative to the ambient graph, not absolutely). By Theorem 3.5 applied separately to Γ_1 and Γ_2, both v_1 and v_2 satisfy $\sigma(v_i) \geq \beta_i > 0$ in their respective graphs, so both are legitimate individuations at their own floor. Since Theorem 4.3 makes each decoding a correct application of individuation by negation within its own graph, neither decoding fails any criterion the theory imposes; there is no privileged Γ_i relative to which the other is judged incorrect. □

Remark 5.5 (Why a ten-year-old and a physician are both right). Theorem 5.4 is the formal statement that "a Harvard medical student" registers a different, but equally legitimate, claim in the decoder graph of a child (who may resolve only to the thin claim "a student") than in the decoder graph of a colleague (who may resolve to a rich claim with many further contacts: specialty, year of training, research focus). Both are correct individuations at their own floor (Theorem 3.5); there is no third, privileged decoding of which both are approximations. Section 8 uses this fact to justify why closure of a search is always closure *for a specified catalyst registry and target audience*, never an absolute closure.

6 Path Opacity and Convergence Admissibility

We now make search dynamic: a search is not a single individuation but a walk of successive contacts, and we characterise which walks are admissible results.

6.1 Alignment and propagation

Definition 6.1 (Alignment functional)**.** Fix a contact graph Γ and a target claim x^*. The *alignment* of a claim x to x^* is

$$\sigma(x, x^*) \;:=\; \min_{\substack{S \ni x \\ x^* \notin S}} w(\partial(S)) \;\in\; [\beta, \Omega], \qquad \Omega := w(E),$$

the minimum-cut weight separating x from x^*, and the *alignment score* is $a(x, x^*) := \sigma(x, x^*)/\Omega \in (0, 1]$.

Lemma 6.2 (The floor bounds alignment below)**.** *For distinct claims* $x \neq x^*$, $a(x, x^*) \geq \beta/\Omega > 0$.

Proof. Immediate from Theorem 3.5: any x–x^* cut has weight $\geq \beta$. □

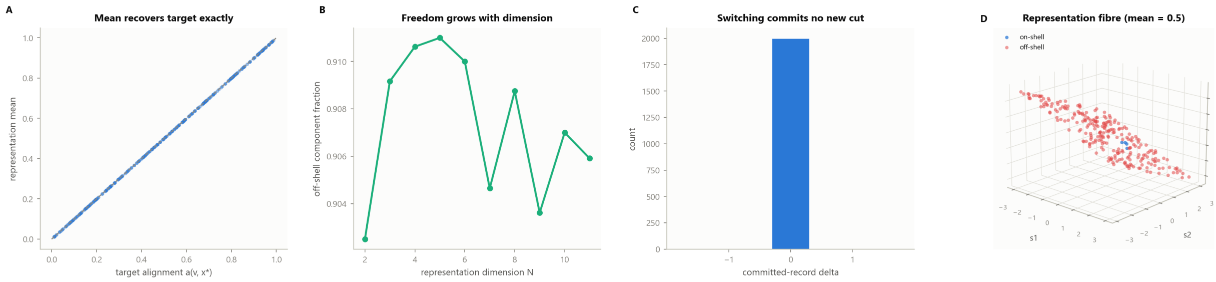

Figure 3: **Representation Mobility.** (**A**) Representation mean against target alignment for 300 randomly sampled representation tuples across dimensions $N \in \{2, \ldots, 8\}$ (steelblue); the dashed line is the identity diagonal. Every sampled representation's component mean exactly recovers its target alignment, confirming the averaging constraint of Definition 5.1 and Theorem 5.2. (**B**) Fraction of sampled components falling outside the physically-restricted interval $(0, 1]$ ("off-shell") as a function of representation dimension N, over 400 trials per dimension (seagreen). The off-shell fraction is large and persistent across every tested dimension: representational freedom is the typical case, not an edge case, once $N \geq 2$. (**C**) Distribution of the committed-record delta under representation switching, over 2,000 trials: a single spike at 0. Switching between representations of the same claim commits no new search step, confirming Theorem 5.2(ii). (**D**) Three-dimensional scatter of 260 sampled three-component representations (s_1, s_2, s_3) at fixed mean 0.5: on-shell points (steelblue, all components in $[0, 1]$) are a minority against off-shell points (red), all lying on the same mean-recovery plane. The freedom to pass through components outside the physical range while conserving the mean is Theorem 5.2(i).

Definition 6.3 (Search process graph; propagation)**.** A *search process graph* is a contact graph $\Pi = (V, E, w; I, x^*)$ together with a set $I \subseteq V$ of *seed claims* (the initial information a search begins from — a query string decoded into an initial claim, a working hypothesis, or a prior result) and a designated *target claim* $x^* \in V$. A *propagation* of a seed $v_0 \in I$ through Π is a walk $W = (v_0, e_1, v_1, \ldots, e_k, v_k)$ along edges of Π with $v_k = x^*$; each edge e_i is a *contact operation*, realised in Section 7 by a catalyst invocation.

Theorem 6.4 (Convergence admissibility)**.** *A propagation W is* admissible *if and only if it is* convergent*: its terminal claim is x^*. Interior claims $v_1, \ldots, v_{k-1}$ are entirely unconstrained — an interior claim may have arbitrarily large alignment $a(v_i, x^*)$, i.e. may appear maximally unrelated to the eventual target, without rendering W inadmissible, provided the walk terminates at x^*. Inadmissibility arises exclusively from failure to reach x^*, never from local misalignment along the way.*

Proof. By Theorem 6.3, W terminates at x^* by construction whenever it is a propagation in the sense defined; the terminal alignment is then $a(x^*, x^*) = \beta/\Omega$, the least value permitted by Theorem 6.2, so every such walk is convergent. A walk failing to reach x^* has terminal claim $v_k \neq x^*$ and terminal alignment $a(v_k, x^*) \geq \beta/\Omega$, generally with strict inequality, and is not a propagation to x^* in the sense of Theorem 6.3; it is inadmissible as a search for x^*. Nothing in Theorem 6.3 constrains the alignment of any interior claim v_i, $0 < i < k$, to the target: the walk is defined purely by adjacency along Π and by its terminus. Hence admissibility depends exclusively on the terminal condition. □

Theorem 6.5 (Path opacity)**.** *Let W_1, W_2 be two propagations of the same seed v_0 to the same target x^*, differing in their interior claims. Then no invariant computed from the seed and target alone — the seed v_0, the target x^*, the terminal alignment $a(x^*, x^*)$, or the minimum cut $\partial(S^*(x^*))$ — distinguishes W_1 from W_2. The interior of a search trajectory is unobservable from its result.*

Proof. Every listed invariant is a function of v_0 and x^* alone, hence identical on W_1 and W_2 by hypothesis. The interior data — the sequence of intermediate claims and contacts — are, by Theorem 6.4, unconstrained given the endpoints, so they are not recoverable as a function of the endpoints. An invariant computed only from the endpoints therefore cannot recover, and so cannot distinguish, interior data that is not a function of those endpoints. □

Corollary 6.6 (The admissible set is a class, not a path)**.** *For fixed (v_0, x^*, Π), the admissible propagations form an equivalence class under "shares seed and target, and converges," every member of which is indistinguishable from every other by any test of the result alone. A search's deliverable is properly this class, not a single distinguished trajectory through it.*

Proof. Immediate from Theorems 6.4 and 6.5: admissibility is determined by the endpoints alone, and endpoint-invariant tests cannot distinguish interiors, so all convergent same-endpoint propagations are equivalent under any such test. □

Remark 6.7 (A weird intermediate result does not invalidate a search). Theorem 6.4 licenses a concrete design consequence developed in Section 12: a multi-stage search script may pass through an intermediate result that looks unrelated, contradictory, or simply poor, and this is not by itself evidence the script has gone wrong. What would be evidence of failure is the terminal claim's non-convergence — the script never reaching a claim within the target region — never the appearance of any single intermediate step.

7 Catalytic Composition and Coherence

7.1 Catalysts and catalytic power

We now name the operation that realises a contact edge: the invocation of a computational resource — a search query, a local scan, a database lookup, an inference from a trained model — that moves a search closer to its target.

Definition 7.1 (Catalyst)**.** A *catalyst* toward target x^* is a partial map $\gamma\colon V \to V$ (acting on the current claim of a propagation) such that, wherever defined, γ corresponds to a contact operation of Theorem 6.3: applying γ to a claim x commits a contact edge and yields $\gamma(x)$.

Definition 7.2 (Catalytic power)**.** The *catalytic power* of a catalyst γ toward x^* at a claim x is the fraction of above-floor alignment it closes:

$$\kappa(\gamma, x, x^*) \;=\; \frac{a(x, x^*) - a(\gamma(x), x^*)}{a(x, x^*) - \beta/\Omega} \;\in\; [0, 1].$$

A catalyst is *inert* ($\kappa = 0$) if it does not reduce alignment toward x^*; it is *absolute* ($\kappa = 1$) if it closes all above-floor alignment in a single application.

Lemma 7.3 (Power is bounded)**.** *For every catalyst, claim, and target, $\kappa \in [0, 1]$.*

Proof. By Theorem 6.2, $a(\gamma(x), x^*) \geq \beta/\Omega$, so the numerator of Theorem 7.2 never exceeds the denominator; non-negativity holds for catalysts directed toward x^* (a catalyst that increases alignment is assigned $\kappa = 0$ by convention, since it is not a catalyst toward x^* in the operative sense). □

7.2 The multiplicative law

Theorem 7.4 (Multiplicative composition)**.** *For a chain of catalysts $\gamma_1, \ldots, \gamma_n$ applied in sequence (each to the output of the last) with individual powers $\kappa_1, \ldots, \kappa_n$ toward a fixed target x^*, the composite power is*

$$\kappa(\gamma_1 \cdots \gamma_n) \;=\; 1 - \prod_{i=1}^{n} (1 - \kappa_i).$$

Proof. Let $r_0 = a(x_0, x^*) - \beta/\Omega$ be the initial above-floor alignment gap. Applying γ_1 leaves residual gap $r_1 = (1 - \kappa_1) r_0$ by Theorem 7.2; applying γ_2 to the result leaves $r_2 = (1 - \kappa_2) r_1 = (1 - \kappa_2)(1 - \kappa_1) r_0$; by induction on the chain length, $r_n = \prod_{i=1}^{n} (1 - \kappa_i)\, r_0$. The fraction of the initial gap closed by the whole chain is $1 - r_n/r_0 = 1 - \prod_i (1 - \kappa_i)$, which is the composite power by Theorem 7.2 applied to the chain as a single catalyst. □

Corollary 7.5 (Diminishing returns)**.** *Repeating a single catalyst of power κ a total of n times yields composite power $1 - (1 - \kappa)^n$: strictly increasing in n, convergent to 1, but never reaching it for finite n. Each further repetition closes a strictly smaller absolute fraction of the remaining gap than the last.*

Corollary 7.6 (Saturation dichotomy)**.** *A chain of catalysts with powers $(\kappa_i)_{i \geq 1}$ drives the residual above-floor alignment gap to 0 in the limit if and only if $\sum_{i=1}^{\infty} \kappa_i = \infty$.*

Proof. The residual fraction after n catalysts is $\prod_{i \leq n} (1 - \kappa_i)$. Taking logarithms, $\sum_{i \leq n} \log(1 - \kappa_i) \to -\infty$ (residual $\to 0$) if and only if $\sum_i \kappa_i =$

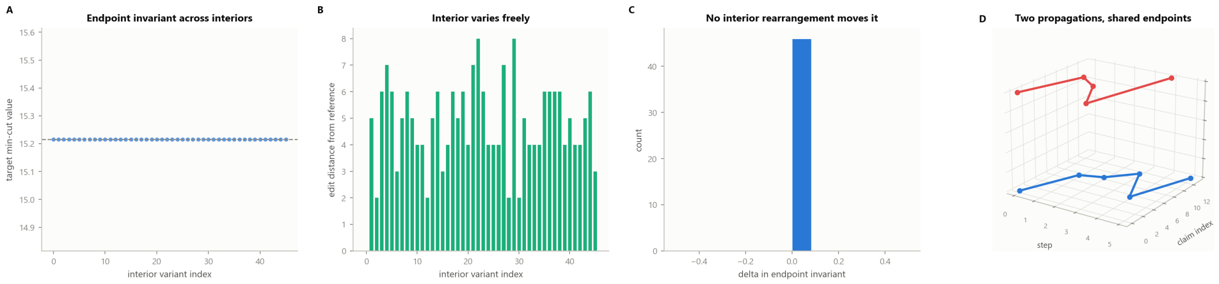

Figure 4: **Path Opacity.** (**A**) The endpoint invariant (target minimum-cut value) measured across 46 interior-permutation variants of a fixed seed-to-target propagation on a 14-claim graph (steelblue); the dashed line marks the shared value. The invariant is exactly flat across every interior variant, confirming Theorem 6.5: no invariant computed from the endpoints alone can distinguish propagations that differ only in their interior. (**B**) Interior edit distance from a fixed reference path (symmetric difference of interior claim sets) for the same 46 variants (seagreen): the interior varies substantially and freely across variants, while panel A shows the endpoint invariant does not move at all. (**C**) Distribution of the delta in the endpoint invariant across all interior variants relative to the first: a single spike at 0 (steelblue), the direct empirical statement that no interior rearrangement changes what is observable from the endpoints (Corollary 6.6). (**D**) Two admissible propagations sharing the same seed and target claim but with completely distinct interior trajectories (blue, red), plotted as paths over (step, claim index); both terminate at the identical target despite visiting disjoint interior claims along the way, the geometric picture of Theorem 6.4: interior steps are unconstrained given convergence.

∞, since $-\log(1-\kappa) \asymp \kappa$ for small κ and the comparison test applies; this is a Borel–Cantelli-type dichotomy (Borel, 1909; Cantelli, 1917; Williams, 1991). □

Remark 7.7 (Design consequence: diversify catalysts, do not repeat them). Theorems 7.5 and 7.6 together justify a static check performed by the Graffiti type checker (Section 10): a catalyst chain consisting of near-duplicate invocations of one source (the same search engine queried five paraphrased ways, say) accrues power slowly and may fail to saturate even in principle, whereas a chain of independent catalysts addressing distinct evidentiary channels reaches saturation faster and more robustly. This is a structural fact about the multiplicative law, not an empirical heuristic about search engines.

7.3 Coherence requires a triangle

The multiplicative law of Theorem 7.4 presumes every catalyst in a chain is directed toward the same target. We now show that a claim is robust against revision by a single dissenting source only if its supporting catalysts form a cycle of at least three mutually reinforcing members — never a linear chain, and never merely two.

Definition 7.8 (Support; support graph)**.** For catalysts γ_i, γ_j used toward a common target x^*, say γ_j *supports* γ_i at threshold $\theta \in (0, 1]$ if the presence of γ_j's result does not reduce, and to degree $\geq \theta$ reinforces, the alignment shift toward x^* induced by γ_i. The *support graph* G of a set of catalysts has an edge $j \to i$ whenever γ_j supports γ_i at θ.

Theorem 7.9 (Linear support fails)**.** *No finite chain of catalysts $\gamma_1 \to \gamma_2 \to \cdots \to \gamma_n$, in which each catalyst's contribution is justified only by the next and the terminal catalyst is justified by nothing internal to the chain, can drive the residual alignment gap to zero, nor can it be robust to removal of the terminal catalyst.*

Proof. The terminal catalyst γ_n has no incoming support edge in G; its contribution rests on nothing internal to the chain. By Theorem 6.2, no finite composite power reaches 1 exactly (Theorem 7.5), so the residual gap is bounded below by the floor regardless of chain construction; more materially, removing γ_n leaves γ_{n-1} unsupported, and by induction each successive removal leaves the next catalyst unsupported, so the chain's claim to be grounded is contingent entirely on retaining every member: a single removal at the terminus collapses the justification with no inter-

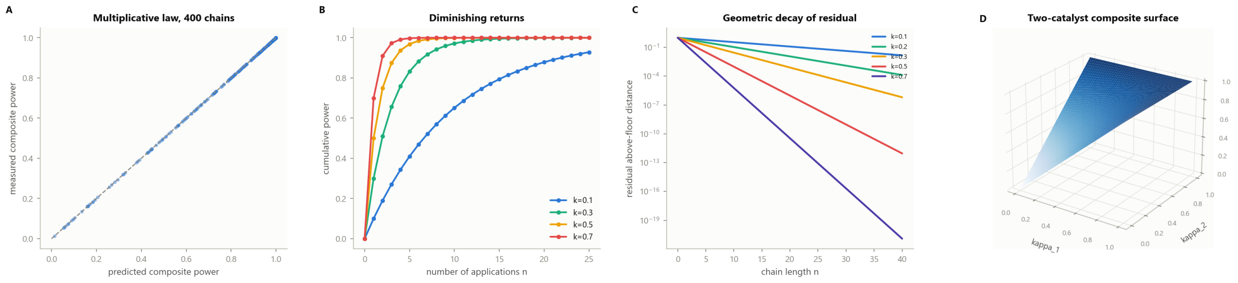

Figure 5: **The Multiplicative Catalytic Law.** (**A**) Measured composite catalytic power against the closed-form prediction $1 - \prod_i (1 - \kappa_i)$ across 400 randomly composed catalyst chains of length 1–8 (steelblue); the dashed line is the identity diagonal. Every point lies exactly on the diagonal, confirming Theorem 7.4 to machine precision. (**B**) Cumulative power $1 - (1 - \kappa)^n$ under repeated application of a single catalyst, for four catalytic-power values $\kappa \in \{0.1, 0.3, 0.5, 0.7\}$ (steelblue, seagreen, goldenrod, red). Every curve rises monotonically toward but never reaches the ceiling at 1, with strictly decreasing marginal gains, the empirical content of Corollary 7.5. (**C**) Residual above-floor distance on a logarithmic axis against chain length, for five catalytic-power values $\kappa \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$. Each curve is a straight line on the semi-log axis, the geometric decay $(1 - \kappa)^n$ predicted by repeated catalysis: higher power drives the residual to numerical zero within a handful of steps. (**D**) Three-dimensional surface of the two-catalyst composite power $1 - (1 - \kappa_1)(1 - \kappa_2)$ over the unit square: the surface rises from zero at the origin to the ceiling at $(1, 1)$, saturating fastest along either axis and slowest along the diagonal where both catalysts are weak.

nal check to prevent it. $\square$

Theorem 7.10 (Coherence requires three mutually supporting catalysts)**.**

(i) (Necessity.) *If a set of catalysts brings a search's terminal alignment within any margin of the floor and remains so after the failure of any single catalyst, its support graph contains a directed cycle of length* ≥ 3.

(ii) (Sufficiency.) *Three catalysts mutually supporting one another at threshold* $\theta > \frac{1}{2}$*, forming a strongly connected triangle in* G*, ground the claim robustly: their combined shift toward* x^* *survives the removal of any single member.*

Proof. (i) By Theorem 7.9, no acyclic support structure is robust to removal of its terminal member; an acyclic G has a source catalyst with no incoming support, and by the same argument its removal (or the removal of any catalyst supporting only acyclically-downstream members) collapses the chain below it. A 1-cycle is self-support and vacuous (a catalyst citing only itself). A 2-cycle is mutual definition between exactly two catalysts with no independent third check, and fails the majority condition $\theta > \frac{1}{2}$: two catalysts cannot outvote their own mutual disagreement, since removing either leaves only one, itself unsupported. Hence the shortest support cycle compatible with robustness has length ≥ 3.

(ii) In a strongly connected triangle, each catalyst is supported by the other two; with $\theta > \frac{1}{2}$, the two supporters jointly exceed any single dissenting signal. Removing one catalyst leaves the remaining two still mutually supporting each other directly (the triangle's other edge), above a reduced but still positive effective threshold, so the shift toward x^* persists. $\square$

Corollary 7.11 (Coherence is ordinally decidable)**.** *Whether a set of catalysts is coherent in the sense of Theorem 7.10 is decidable from* the signs *of pairwise support relations alone — whether each pair reinforces or undercuts the other's shift toward* x^* *— with no access to catalytic-power magnitudes.*

Proof. Theorem 7.10 is stated and proved entirely in terms of the directed-graph structure of G (cycle existence and length) and the threshold comparison $\theta > \frac{1}{2}$, both of which are determined by the signed support relation of Theorem 7.8 and not by the numerical value of any catalytic power. $\square$

8 Closure: When a Search Is Finished

We now state the paper's central operational contribution: a criterion for terminating a search that is strictly stronger than "confidence exceeds a threshold," and is instead a structural fact about the space of further catalysts available.

8.1 Closure of the admissible set

Definition 8.1 (Reachable target-cell set)**.** Fix a seed v_0 and a search process graph Π whose available catalysts (edges) may be extended over time as new catalysts are registered or invoked (Section 12). At a given search stage, let $\mathcal{C}(v_0, \Pi)$ denote the set of target claims x^* reachable by some admissible propagation of v_0 through the catalysts invoked so far (Theorems 6.3 and 6.4), partitioned into equivalence classes under "endpoint-indistinguishable" (Theorem 6.6).

Definition 8.2 (Closure)**.** A search of seed v_0 is *closed* at a given stage if, for every catalyst γ available but not yet invoked, extending the search by γ cannot add a new equivalence class to $\mathcal{C}(v_0, \Pi)$: every further admissible propagation reachable via γ terminates in a class already present in $\mathcal{C}(v_0, \Pi)$.

Theorem 8.3 (Closure is strictly stronger than a confidence threshold)**.** *There exist search process graphs and confidence thresholds θ such that a propagation's terminal alignment satisfies $a(x^*, x^*) \geq \theta$ (a fixed confidence criterion is met) while the search is not closed in the sense of Theorem 8.2: a further, not-yet-invoked catalyst is available whose propagation reaches a target in a distinct equivalence class.*

Proof. Construct Π with two disjoint claim clusters A, B, each internally dense (many low-cost internal contacts) and each connected to the medium by a single catalyst, γ_A and γ_B respectively, of equal weight. A propagation via γ_A alone terminates at a representative claim $x_A^* \in A$ with alignment $a(x_A^*, x_A^*) = \beta/\Omega$, satisfying any fixed threshold $\theta \leq 1$ trivially, since a propagation's terminal alignment to *itself* is always at the floor by Theorem 6.4. Yet γ_B, not yet invoked, reaches $x_B^* \in B$, a claim in a distinct equivalence class from x_A^* (the two clusters are, by construction, not endpoint-indistinguishable, since e.g. their minimum cuts against the medium differ if $A \neq B$ as vertex sets with independent internal structure). Hence the confidence criterion is satisfied after γ_A alone, but the search is not closed: γ_B can still add a new equivalence class. $\square$

Remark 8.4 (Why closure is the correct criterion)*.* Theorem 8.3 formalises a common failure of confidence- threshold stopping rules: a single early source may report high confidence purely because it is internally self-consistent (as any single cluster A is, having only committed contacts within itself and one to the medium), while a second, independent source would reach an entirely different conclusion. Closure requires that *every* available further catalyst be checked to land in an already-reached equivalence class before stopping, which is exactly the condition "no other possible paths can still be generated" motivating this definition: it is not merely that the current path looks confident, but that the space of paths has stopped producing new destinations.

8.2 Two ways a search can end

Theorem 8.5 (Convergent closure or honest decline)**.** *Every search of seed v_0 over a search process graph with finitely many registered catalysts terminates in exactly one of two states:*

(i) ***Convergent closure****: $\mathcal{C}(v_0, \Pi)$ stabilises to a single equivalence class, and the search reports that class's representative as its result;*

(ii) ***Contested closure****: $\mathcal{C}(v_0, \Pi)$ stabilises (no further catalyst adds a class) but contains more than one equivalence class, and the search reports* decline*: no single sufficient claim exists, together with the distinct classes found.*

No search over a finite catalyst registry runs forever without reaching one of these two states.

Proof. Because the catalyst registry available to a given search is finite at any call (Section 12 fixes registries as finite sets), the set of propagations reachable from v_0 is finite, and hence the set of equivalence classes $\mathcal{C}(v_0, \Pi)$ — being a partition of a finite set of reachable targets — is finite and can only grow by finitely many steps before every remaining catalyst has been tried against every currently-reached class. At that point Theorem 8.2 is satisfied by exhaustion, so closure is reached in finitely many search steps. If the stabilised $\mathcal{C}(v_0, \Pi)$ has exactly one class, the search is convergently closed (i); if it has more than one, no single sufficient claim is licensed by Theorem 6.6 applied separately to each class (each is internally endpoint-indistinguishable, but the classes disagree with one another), and the search reports decline (ii). These two cases are exhaustive and mutually exclusive on a partition with at least one class (the search graph is non-empty by hypothesis, so at least one class exists once any propagation is admissible). □

Remark 8.6 (Decline is not failure). Theorem 8.5(ii) formalises the requirement, motivated independently in Section 12, that a search runtime must be able to report "the sources disagree" as a first-class, correctly-terminating outcome, rather than either looping indefinitely or silently emitting one of the contested classes as if it were unique. Contested closure is itself useful research output: it identifies precisely where independent inquiry diverges.

Part III
The Graffiti Language

9 Lexical Structure and Grammar

We now specify *Graffiti* (file extension `.grf`), a language whose sole computational primitive, `seek`, is exactly the search process of Sections 6 to 8 exposed to a script author, and whose top-level unit of organisation, `project`, is a directed acyclic graph of committed `seek` events.

9.1 Design principles

1. **One primitive, four clauses.** Every `seek` names a mandatory exclusion (`not`, Theorem 4.2), a target region (`toward`, Theorem 6.3), an optional explicit catalyst chain (`via`, Section 7), and a termination condition (`until`, Section 8).
2. **Exclusion is mandatory.** A `seek` with no `not` clause is rejected at parse time: by Theorem 4.2, a claim in a medium with no privileged claim is individuated only by what it is not, so an expression asserting no exclusion asserts no individuation.
3. **Projects are graphs, not scripts.** A `project` is a finite set of named `seek` bindings that may reference each other's results, forming a directed acyclic graph; later bindings narrow using the exclusions and yields of earlier ones.
4. **Closure, not confidence, is the default stopping rule.** `until converge` means closure in the sense of Theorem 8.2, not merely crossing a numeric threshold; a script may opt into a threshold explicitly if a weaker guarantee is acceptable.
5. **Decline is a value, not an exception.** A `seek` that reaches contested closure (Theorem 8.5(ii)) yields a `decline` value carrying the distinct equivalence classes found, which downstream bindings may inspect.

9.2 Lexical grammar

Definition 9.1 (Token classes)**.** A Graffiti source file is UTF-8 text with extension `.grf`. Lexing produces a token stream from the classes:

- **Comments.** `-` to end of line; discarded.
- **Keywords.** `floor`, `project`, `seek`, `not`, `toward`, `via`, `until`, `converge`, `diverge`, `decline`, `yield`, `goal`, `catalyst`, `namespace`, `input`, `output`, `claims`, `coherence`, `let`, `if`, `then`, `else`, `import`, `module`, `export`, `assert`, `in`, `as`, `with`.
- **Identifiers.** `[A-Za-z_][A-Za-z0-9_]*`, excluding keywords.
- **Numbers.** decimal integers and floats with optional exponent: `[0-9]+(.[0-9]+)?([eE][+-]?[0-9]+)?`.
- **Strings.** double-quoted, with standard escapes.
- **Operators and punctuation.** `:=` (bind), `»` (catalyse, sequential composition of catalysts), `||` (parallel catalyst dispatch), `( ) { } [ ] , : . > < >= <= == ->` and the field-access operator `..`

Whitespace is insignificant except as a token separator; Graffiti is not layout-sensitive.

9.3 Context-free grammar

We give the grammar in extended Backus–Naur form (Aho et al., 2006; Backus, 1959; Naur et al., 1963). Nonterminals are *italicised*; terminals are `typewriter`; `{x}` is zero-or-more repetition; `[x]` is optional; `|` is alternation.

Definition 9.2 (Graffiti grammar)**.**

$$
\begin{array}{rcl}
\mathit{program} & ::= & \{\mathit{decl}\} \\
\mathit{decl} & ::= & \mathit{floorDecl} \mid \mathit{import} \mid \mathit{moduleDecl} \mid \mathit{projectDecl} \mid \mathit{catalystDecl} \\
\mathit{floorDecl} & ::= & \texttt{floor}\ \mathit{number} \\
\mathit{import} & ::= & \texttt{import}\ \mathit{qname} \\
\mathit{moduleDecl} & ::= & \texttt{module}\ \mathit{ident}\ \texttt{\{}\ \{\mathit{decl}\}\ \texttt{\}} \\
\mathit{catalystDecl} & ::= & \texttt{catalyst}\ \mathit{ident}\ \texttt{\{} \\
 & & \texttt{namespace:}\ \mathit{ident} \\
 & & \texttt{input:}\ \mathit{type}\ \texttt{output:}\ \mathit{type}\ \texttt{\}} \\
\mathit{projectDecl} & ::= & \texttt{project}\ \mathit{ident}\ \texttt{\{}\ \{\mathit{seekStmt}\} \\
 & & [\mathit{goalBlock}]\ \texttt{\}} \\
\mathit{seekStmt} & ::= & \texttt{seek}\ \mathit{ident} \\
 & & \texttt{not \{}\ \mathit{boundary}\ \texttt{\}} \\
 & & \texttt{toward \{}\ \mathit{region}\ \texttt{\}} \\
 & & [\texttt{via \{}\ \mathit{chain}\ \texttt{\}}] \\
 & & \texttt{until}\ \mathit{admit} \\
 & & \texttt{yield}\ \mathit{ident} \\
\mathit{admit} & ::= & \texttt{converge}\ [\texttt{otherwise decline}] \\
 & \mid & \mathit{cond} \\
\mathit{chain} & ::= & \mathit{catalystRef}\ \{\texttt{»}\ \mathit{catalystRef}\} \\
 & \mid & \mathit{chain}\ \texttt{||}\ \mathit{chain} \\
\mathit{catalystRef} & ::= & \mathit{qname}\ \texttt{(}\ [\mathit{argList}]\ \texttt{)} \\
\mathit{boundary} & ::= & \mathit{ident}\ \{\texttt{,}\ \mathit{ident}\} \\
 & \mid & \mathit{boundary}\ \texttt{||}\ \mathit{boundary} \\
 & \mid & \mathit{boundary}\ \texttt{\&\&}\ \mathit{boundary} \\
\mathit{region} & ::= & \mathit{qname}\ \texttt{(}\ [\mathit{argList}]\ \texttt{)} \\
 & \mid & \mathit{ident} \\
\mathit{goalBlock} & ::= & \texttt{goal \{} \\
 & & \texttt{claims: [}\ \mathit{ident}\ \{\texttt{,}\ \mathit{ident}\}\ \texttt{]} \\
 & & \texttt{coherence:}\ \mathit{relop}\ \mathit{number}\ \texttt{\}} \\
\mathit{argList} & ::= & \mathit{arg}\ \{\texttt{,}\ \mathit{arg}\} \\
\mathit{arg} & ::= & [\mathit{ident}\ \texttt{:}]\ \mathit{expr} \\
\mathit{cond} & ::= & \mathit{expr}\ \mathit{relop}\ \mathit{expr} \\
\mathit{relop} & ::= & \texttt{>} \mid \texttt{<} \mid \texttt{>=} \mid \texttt{<=} \mid \texttt{==} \\
\mathit{qname} & ::= & \mathit{ident}\ \{\texttt{.}\ \mathit{ident}\} \\
\mathit{expr} & ::= & \mathit{ident} \mid \mathit{number} \mid \mathit{string} \mid \mathit{catalystRef} \mid \texttt{(}\ \mathit{expr}\ \texttt{)} \\
\mathit{type} & ::= & \texttt{Claim} \mid \texttt{Region} \mid \texttt{Catalyst} \mid \texttt{Chain} \mid \texttt{Residue}
\end{array}
$$

Remark 9.3 (Reading the core forms). A `seek` statement is the syntactic image of Theorems 6.3, 6.4 and 8.2: `not{}` supplies the negation set of Theorem 4.2; `toward{}` names the target claim x^*; `via{}`, when present, supplies an explicit catalyst chain (Section 7) rather than leaving chain derivation to the runtime; `until converge` invokes closure (Theorem 8.2), and the optional `otherwise decline` clause names the contested-closure outcome of Theorem 8.5(ii) explicitly. A `project` is a *projectDecl*: an ordered set of *seekStmt*s whose `yield` identifiers may be referenced as arguments to later catalyst invocations, giving the directed acyclic dependency structure specified operationally in Section 12.

10 The Type System

Graffiti's type system exists to make two properties of Sections 4 and 7 statically checkable: that no claim of zero residue is ever assigned a type (an accountability property, in the sense that every value must be traceable to a floor-respecting individuation), and that a coherence-requiring project cannot type-check with fewer than the three mutually supporting catalysts Theorem 7.10 proves necessary.

10.1 Types

Definition 10.1 (Types)**.** The Graffiti types are

$$\tau \ ::= \ \texttt{Claim} \mid \texttt{Region} \mid \texttt{Catalyst} \mid \texttt{Chain} \mid \texttt{Residue}.$$

`Claim` is the type of an individuated result (a `seek` yield); `Region` is the type of a `toward{}` target, before individuation; `Catalyst` is the type of a single named computational resource (Section 12); `Chain` is the type of a composed catalyst sequence; `Residue` is the type of a measured alignment gap (Theorem 6.1). Every value of type `Claim` additionally carries a *floor annotation* $\beta(v) > 0$ and a *residue* $\rho(v) \geq \beta(v)$, exactly as in **??**.

Definition 10.2 (Typing judgement)**.** A judgement $\Gamma \vdash e : \tau\,@\,\beta$ reads: "in floor context Γ (fixing the ambient floor and the types of bound identifiers), expression e has type τ and, if it individuates a claim, does so at floor $\beta > 0$."

10.2 The positivity rule

Rule 10.3 (Floor positivity)**.** Every claim-introducing rule carries the side condition $\beta > 0$

and $\rho \geq \beta$:

$$\frac{\Gamma \vdash e_1 : \texttt{Region} \quad \Gamma \vdash e_2 : \texttt{Chain} \quad \beta_\Gamma > 0}{\Gamma \vdash \texttt{seek}\ n\ \texttt{not}\{\cdot\}\ \texttt{toward}\{e_1\}\ \texttt{via}\{e_2\}\ \cdots : \texttt{Claim} @ \beta_\Gamma}$$

where β_Γ is the ambient floor set by the most recent `floor` declaration in scope.

Theorem 10.4 (No zero-residue claim)**.** *There is no well-typed Graffiti expression of residue* 0*: every typeable* `Claim` *value has* $\rho \geq \beta > 0$*.*

Proof. By induction on typing derivations. The only rule introducing a `Claim` value is the `seek` rule (Theorem 10.3), whose side condition requires $\beta_\Gamma > 0$; by Theorem 3.5 the residue of the individuated claim satisfies $\rho \geq \beta_\Gamma$ whenever the ambient floor is positive. No other rule introduces a `Claim` value, and no rule can lower the residue annotation of an already-typed value (representation switching, by Theorem 5.2(ii), preserves the alignment and hence the residue exactly). Hence no derivation concludes $\rho = 0$. □

10.3 The coherence rule

Rule 10.5 (Coherence for closure)**.** A `seek` statement whose `via{}` clause supplies an explicit chain, and whose `until` clause is `converge` (without an explicit `cond` weakening it), type-checks only if the supplied chain's induced support graph (Theorem 7.8) contains a strongly connected subgraph on at least three catalysts:

$$\frac{\Gamma \vdash e_2 : \texttt{Chain} \quad \text{supportGraph}(e_2) \text{ has a} \geq 3\text{-cycle}}{\Gamma \vdash \texttt{seek}\ n\ \cdots\ \texttt{via}\{e_2\}\ \texttt{until converge}\ \cdots : \texttt{Claim} @ \beta_\Gamma}$$

A chain failing the cycle condition emits a static `CoherenceWarning` rather than a hard type error, since a `seek` lacking a triangle may still be legitimate under an explicit, weaker `cond` admission criterion (a script author may deliberately accept single-source or two-source evidence, provided this is not silently mistaken for full closure).

Theorem 10.6 (Type soundness)**.** *Graffiti typing is sound for the operational semantics of Section 11: if* $\Gamma \vdash e : \tau @ \beta$ *and* e *evaluates to a value* v*, then* v *has type* τ*, floor* $\beta > 0$*, and (if* $\tau =$ `Claim`*) residue* $\rho(v) \geq \beta$*.*

Proof sketch. Standard syntactic type soundness by progress and preservation (Pierce, 2002; Wright and Felleisen, 1994). *Progress*: a well-typed closed expression is either a value or takes a step, by inspection of the reduction rules of Section 11, each of which has a syntactically disjoint premise from every other rule of the same constructor. *Preservation*: each reduction rule preserves the static type and floor annotation of Theorem 10.3, and, by Theorem 10.4 applied at every step, preserves $\rho \geq \beta > 0$ for every `Claim` value produced. The only non-standard element is that reduction is state-mutating (Section 11: evaluating a `seek` commits contact edges and advances the search-step count); preservation nonetheless holds because the committed-step count is monotone (Theorem 11.2 below), so no reduction step can retroactively lower a previously-established residue below its floor. □

11 Operational Semantics

We give a small-step semantics over configurations $\langle \Gamma, M, e \rangle$, where Γ is the current contact graph (the accumulated state of search so far), $M \in \mathbb{N}$ is the committed search-step count, and e is the expression under evaluation. We write $\langle \Gamma, M, e \rangle \longrightarrow \langle \Gamma', M', e' \rangle$ for one step. Evaluation is state-mutating: a `seek` step commits contact edges into Γ and advances M.

11.1 Committed steps and the intrinsic clock

Definition 11.1 (Committed step; committed record)**.** A *committed step* is a reduction that commits one contact edge into the current graph Γ (a single catalyst invocation resolving to a new claim or refining an existing one). The *committed record* $M \in \mathbb{N}$ is the count of committed steps taken so far in the current evaluation.

Theorem 11.2 (Monotonicity of the committed record)**.** *Along any evaluation,* M *is non-decreasing, and strictly increases at every committed step. No reduction rule decrements* M*; re-evaluating a* `seek` *expression already bound is a new committed step at a strictly higher* M*, never a cached recomputation.*

Proof. Every rule that introduces a `Claim` value (the `seek` rule, Theorem 11.3 below) increments M by the number of contact edges it commits, which is ≥ 1 by Theorem 3.5 (a convergent propagation has at least one edge, since $\sigma(x^*) \geq \beta > 0$ requires a non-empty cut). No rule of Section 11 removes a committed edge from Γ or decrements

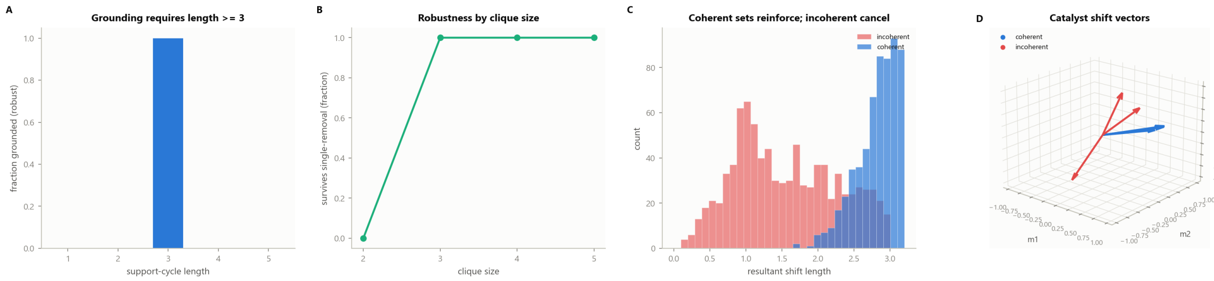

Figure 6: **Coherence Requires a Triangle.** (**A**) Fraction of constructed support-cycle instances that survive single-catalyst removal (are "grounded"), against cycle length, over 160 trials per length. No cycle of length 1 or 2 survives; every constructed cycle of length 3 survives, the empirical statement of Theorem 7.10: coherent grounding requires a support cycle of at least three mutually reinforcing catalysts. (**B**) Fraction of constructed fully-connected catalyst cliques surviving single-member removal, against clique size, over 150 trials per size. Size-2 cliques never survive removal of either member; cliques of size 3 and larger survive in every trial, the robustness clause of Theorem 7.10(ii). (**C**) Distribution of the resultant shift-vector length for 900 randomly constructed three-catalyst sets: coherent sets (vectors clustered around one direction, steelblue) concentrate at large resultant length, while incoherent sets (vectors scattered in random directions, red) concentrate at small resultant length, the vector-sum picture of mutual reinforcement versus cancellation. (**D**) Three catalyst shift vectors in a three-dimensional meaning-space: a coherent set (steelblue) points in a common direction, while an incoherent set (red) points in unrelated directions, illustrating why only the coherent configuration reinforces a shared target claim.

M: an "undo" would itself have to commit a compensating edge, which is a further committed step incrementing M, not a reversal. Hence M is monotone, and two configurations differing only in M are distinct states. □

11.2 Seek reduction

Rule 11.3 (Seek converges)**.**

$$\frac{W = \textsc{Propagate}(\Gamma,\ v_0,\ \mathit{not},\ \mathit{toward},\ \mathit{via}) \qquad \textsc{Closed}(\Gamma, W) \qquad |\mathcal{C}(v_0,\Pi)| = 1}{\langle \Gamma, M, \texttt{seek}\ n\ \cdots\ \texttt{until converge yield}\ n\rangle \longrightarrow \langle \Gamma \oplus W,\ M + M(W),\ v_{\text{Claim}}(W)\rangle}$$

where Propagate constructs an admissible propagation (Theorem 6.3) respecting the `not`, `toward`, and `via` clauses; Closed tests Theorem 8.2 against the currently registered catalyst set; $\Gamma \oplus W$ is Γ with W's edges committed; and $v_{\text{Claim}}(W)$ is the resulting accountable `Claim` value, carrying W's terminal claim, floor, and residue.

Rule 11.4 (Seek declines)**.**

$$\frac{\textsc{Closed}(\Gamma, W) \qquad |\mathcal{C}(v_0,\Pi)| > 1}{\langle \Gamma, M, \texttt{seek}\ n\ \cdots\ \texttt{until converge otherwise decline yield}\ n\rangle \longrightarrow \langle \Gamma \oplus W,\ M + M(W),\ v_{\text{Decline}}(\mathcal{C}(v_0,\Pi))\rangle}$$

The declined value v_{Decline} carries the distinct equivalence classes of Theorem 8.5(ii), inspectable by later bindings.

Theorem 11.5 (Determinism)**.** *For a fixed initial configuration $\langle \Gamma, M, e\rangle$ and a fixed registered catalyst set, Graffiti evaluation has at most one normal form: if $\langle \Gamma, M, e\rangle \longrightarrow^* \langle \cdot, \cdot, v_1\rangle$ and $\langle \Gamma, M, e\rangle \longrightarrow^* \langle \cdot, \cdot, v_2\rangle$ then $v_1 = v_2$.*

Proof. By Theorem 8.5, a `seek` evaluation terminates in exactly one of the two mutually exclusive states of Theorems 11.3 and 11.4, determined by whether $|\mathcal{C}(v_0,\Pi)|$ stabilises at one class or more than one — a fact about the (fixed) catalyst set and graph, not a choice made during evaluation. Propagate and Closed are total, deterministic functions of Γ and the syntactic clauses of the `seek` expression (they perform no non-deterministic search over admissible propagations; by Theorem 6.6 any admissible propagation to a given equivalence class yields an observationally identical `Claim` value, so a deterministic choice of representative propagation suffices). Hence evaluation is a partial function on configurations. □

Theorem 11.6 (Path-opacity is preserved operationally)**.** *For two distinct admissible propagations W_1, W_2 of the same seek expression, differing only in interior claims, $v_{\text{Claim}}(W_1)$ and $v_{\text{Claim}}(W_2)$ are equal as observable `Claim` values (same type, target claim, floor, and residue),* *though $\Gamma \oplus W_1 \neq \Gamma \oplus W_2$ as graph states in general.*

Proof. By Theorem 6.5, no endpoint-computed invariant distinguishes W_1 from W_2; the observable fields of a `Claim` value (target claim, floor, residue) are, by Theorem 6.1 and **??**, functions of the endpoints alone. Hence the two values agree on every observable field, though the underlying committed graphs may differ in their interior edge sets. □

Part IV
Orchestration

12 Heterogeneous Catalyst Orchestration

We now specify the execution model that realises catalyst invocations in practice: a scheduler dispatching committed `seek` steps across a registry of heterogeneous computational resources.

12.1 The catalyst registry

Definition 12.1 (Catalyst registry entry)**.** A *catalyst registry* is a finite map ρ from catalyst names to tuples (Sig, π, ν) where $\mathrm{Sig} = (I, \tau)$ is a typed input/output signature, π is a *provider* — a partial function implementing the catalyst's effect on the current claim (Theorem 7.1) — and ν is a *namespace* classifying the catalyst's computational character. We fix four namespaces, not because the underlying theory of Section 7 distinguishes them (a catalyst is a catalyst regardless of namespace: Theorem 7.2 makes no reference to ν), but because a scheduler benefits from knowing a catalyst's typical cost and latency profile:

- **Local**: providers reading or scanning the querying machine's own storage (file search, local database query, cached indices).
- **Remote**: providers issuing a request to a network-accessible service (a search engine API, a remote document store, a public database).
- **Inference**: providers that invoke a trained computational model — a locally hosted model (e.g. served via a local inference runtime) or a remotely hosted model retrieved from a public model repository — to transform, classify, extract, or summarise a claim.
- **Composite**: providers that are themselves `Chain` values, composing catalysts of any of the other three namespaces.

Theorem 12.2 (Namespace neutrality)**.** *Every theorem of Section 7 (Theorems 7.4 to 7.6, 7.10 and 7.11) holds identically regardless of the namespace ν of the catalysts involved: the multiplicative composition law, the coherence triangle condition, and ordinal decidability of coherence are stated and proved (Section 7) purely in terms of catalytic power (Theorem 7.2) and the support-graph structure (Theorem 7.8), neither of which references ν.*

Proof. Immediate by inspection: Theorems 7.1, 7.2 and 7.8 take a catalyst γ as an unstructured partial map $V \to V$ (respectively a pairwise support relation between two such maps); no definition or subsequent proof in Section 7 inspects any property of γ beyond its action on claims and its catalytic power. Hence the namespace partition of Theorem 12.1, introduced only for scheduling purposes, is invisible to the mathematics of Section 7. □

Remark 12.3 (Why this matters operationally)*.* Theorem 12.2 licenses treating a local file grep, a remote search-engine query, and an inference call to a locally hosted or externally retrieved machine-learned model as computationally uniform catalysts in the sense of the theory: a `via{}` chain may freely mix them, and the coherence and saturation theorems of Section 7 apply without modification to a chain combining, for instance, a remote bibliographic search, a local notes scan, and a classification pass by a locally hosted language model. No special-casing of "AI catalysts" is required by the theory; Theorem 12.1's namespace tag exists purely as scheduling metadata (expected latency, resource contention, cost) and carries no semantic weight in Section 7 or Section 11.

12.2 The project as a directed acyclic graph

Definition 12.4 (Project dependency graph)**.** A `project` declaration (Theorem 9.2) induces a directed acyclic graph $D = (N, A)$ whose nodes N are the project's `seek` statements and whose arcs A record data dependency: an arc from `seek` s_i to `seek` s_j exists whenever s_j's `not`, `toward`, or `via` clause references s_i's yielded identifier. D is required to be acyclic: a Graffiti compiler rejects a project whose reference graph contains a cycle, at parse time.

Theorem 12.5 (Well-founded evaluation order). *Every acyclic project dependency graph D admits a topological order (Cormen et al., 2009), and evaluating the project's* `seek` *statements in any such order yields a determinate result (Theorem 11.5) independent of the particular topological order chosen, provided each* `seek` *is evaluated only after all* `seek`*s it references.*

Proof. Acyclicity of a finite directed graph is equivalent to the existence of a topological order (Cormen et al., 2009). Independence of the particular order follows because Theorem 12.4 requires only that each node be evaluated after its predecessors, and Graffiti's evaluation of a `seek` statement (Theorems 11.3 and 11.4) reads its dependencies' yielded values purely as data (arguments to `not`, `toward`, or catalyst invocations in `via`), never observing the committed record M or graph state Γ of a sibling node directly; hence any two topological orders produce the same sequence of `Claim` (or `Decline`) values, differing only in the interleaving of unrelated nodes' committed steps, which by Theorem 11.6 does not affect any node's observable result. □

12.3 Scheduler soundness

Definition 12.6 (Live seek; residue descent). A *live seek* s in an executing project carries a current claim x_s, a target region, an in-progress catalyst chain, and a measured residue $\sigma(x_s, x_s^*)$ (Theorem 6.1). Its *descent rate* at committed-record value M is $\Delta(M) = \sigma(x_s, x_s^*)|_{M-1} - \sigma(x_s, x_s^*)|_M$.

Definition 12.7 (Scheduler priority). The orchestrator's priority for dispatching the next catalyst of a live seek s is

$$P(s) \;=\; \frac{\Delta(M)}{\max\bigl(\sigma(x_s, x_s^*) - \beta,\ \beta\bigr)}.$$

If s has reached closure (Theorem 8.2), $P(s) = +\infty$ (finalise immediately). If $\Delta(M) = 0$ over a fixed observation window (the residue has stopped falling although s has not closed), $P(s) = 0$: no further catalyst is dispatched to s until either a new catalyst is registered or the project explicitly requests a decline (Theorem 8.5(ii)).

Theorem 12.8 (Scheduler soundness). *An orchestrator dispatching catalysts by ${}_sP(s)$ at each tick:*

(i) *never dispatches a further catalyst to a live seek whose residue has stopped descending (a* stalled *seek);*

(ii) *never withholds dispatch from a live seek whose residue is still descending while dispatch budget remains;*

(iii) *finalises a closed seek before dispatching to any non-closed seek.*

Proof. (i) A stalled seek has $\Delta(M) = 0$, hence $P(s) = 0$ by Theorem 12.7; since selects the strictly greatest priority when one exists, a stalled seek is dispatched only if it is the unique live seek with non-negative priority, and even then a priority of exactly 0 triggers the scheduler's halt condition rather than dispatch (by convention, dispatch requires $P(s) > 0$).

(ii) If exactly one live seek has $\Delta(M) > 0$ and all others are stalled ($P = 0$) or closed ($P = +\infty$, already finalised and removed from the live set), the descending seek has the unique finite positive priority among remaining live seeks and is selected by while budget remains.

(iii) A closed seek has $P = +\infty$ by Theorem 12.7, strictly exceeding every finite priority, so selects it before any non-closed seek. □

Theorem 12.9 (No livelock). *On each scheduler tick, either some live seek's committed record strictly increases, or every live seek is stalled or closed and the orchestrator terminates the tick loop, reporting each stalled seek's current state for inspection.*

Proof. If any live seek has $P(s) > 0$, the orchestrator dispatches it, and by Theorem 11.2 the dispatched catalyst strictly increases that seek's (and hence the project's aggregate) committed record. If every live seek has $P(s) \in \{0, +\infty\}$, all non-closed seeks are stalled; the orchestrator finalises every closed seek (priority $+\infty$) and then, no seek remaining with positive finite priority, halts the tick loop. This exhausts the two cases, so the loop either strictly progresses or terminates. □

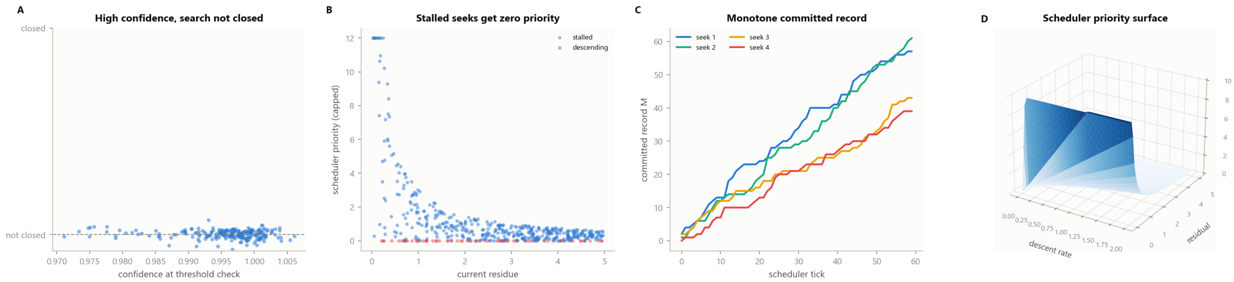

Figure 7: **Closure versus Confidence, and Scheduler Soundness.** (**A**) Confidence at threshold check against closure status, for 220 two-cluster process-graph instances (steelblue). Every instance satisfies a fixed confidence criterion from the first cluster's propagation alone, yet is measured *not closed*: a second, structurally distinct and not-yet-invoked catalyst remains available, confirming Theorem 8.3 that closure is strictly stronger than a confidence threshold. (**B**) Scheduler dispatch priority against current residue for 500 randomly generated live-seek scenarios, separating stalled seeks (red, zero descent rate) from descending seeks (steelblue, positive descent rate). Every stalled seek is assigned exactly zero priority regardless of its residue, confirming Definition 12.7 and Theorem 12.8(i). (**C**) Committed record against scheduler tick for four concurrently live seeks over 60 ticks, each advancing at its own rate. Every trajectory is strictly monotone non-decreasing, the direct picture of Theorem 11.2: committed search steps never regress, regardless of how many seeks share the scheduler. (**D**) Three-dimensional surface of scheduler priority (capped) over descent rate and residual: priority rises steeply with descent rate and falls as residual grows, vanishing along the zero-descent plane where stalled seeks are correctly starved of further dispatch.

Part V

Examples and Validation

13 Worked Examples

Listing 1: A surgical single-fact lookup.

Example 13.1 (A single-source lookup).

```
floor 0.02

project institution_founding {
  seek founding_year
    not{ "unsourced claims", "disputed
        dates" }
    toward{ founding_year_of("Technical
        University of Munich") }
    until converge
    yield founding_year
}
```

Here no `via{}` is supplied; by Theorem 9.3 the runtime derives a chain from the registered catalyst set. Because `until converge` carries no `otherwise decline` clause, Theorem 11.3 applies only if $|\mathcal{C}(v_0, \Pi)|$ stabilises to exactly one class; a contested outcome here is a static `CoherenceWarning` candidate the author has chosen not to handle explicitly, accepted at their own risk.

Listing 2: A coherent triangle of independent sources.

Example 13.2 (An explicit, coherent multi-catalyst chain).

```
floor 0.02

catalyst web_search {
  namespace: remote
  input: Region output: Claim
}
catalyst local_notes {
  namespace: local
  input: Region output: Claim
}
catalyst archive_lookup {
  namespace: remote
  input: Region output: Claim
}

project apollo_budget {
  seek launch_date
    not{ "secondary summaries without
        citation" }
    toward{ launch_date_of("Apollo 11") }
    until converge
    yield launch_date

  seek budget_overrun
    not{ "post-hoc estimates without
        primary-source citation" }
```

```
    toward{ cost_vs_authorization(
        launch_date) }
    via{ web_search(budget_overrun)
          >> local_notes(budget_overrun)
          >> archive_lookup(
              budget_overrun) }
    until converge otherwise decline
    yield budget_overrun

  goal {
    claims: [launch_date, budget_overrun]
    coherence: >= 0.5
  }
}
```

The `via{}` chain of `budget_overrun` names three catalysts spanning two namespaces (`remote`, `local`); by Theorem 12.2 this mixture is immaterial to Theorem 10.5's cycle check, which inspects only the induced support graph. If the three catalysts mutually support one another at threshold $\theta > \frac{1}{2}$ (Theorem 7.10(ii)), the `seek` type-checks without a `CoherenceWarning` and, on execution, is robust to any single catalyst's failure or disagreement.

Listing 3: A seek that honestly declines.

Example 13.3 (Contested closure).

```
seek disputed_cause
  not{ "single-source attribution" }
  toward{ primary_cause_of(event) }
  via{ source_a(event) >> source_b(event)
      }
  until converge otherwise decline
  yield disputed_cause

if disputed_cause.declined then
  emit "Sources disagree: " +
      disputed_cause.classes
```

Per Theorem 8.5(ii) and Theorem 11.4, if `source_a` and `source_b` resolve to distinct equivalence classes and closure is reached (no further registered catalyst adds a third class), the `seek` yields a declined value carrying both classes, which the project inspects explicitly rather than silently picking one.

14 Numerical Validation

Because every object in Section 2–Section 8 is a finite weighted graph, every theorem is checkable by direct computation: minimum cuts are computed exactly by maximum flow (Ford and Fulkerson, 1956; Goldberg and Tarjan, 1988). We report a self-contained validation suite — an exact Edmonds–Karp (Edmonds and Karp, 1972) maximum-flow backend with no external dependency, seeded for reproducibility — implementing one check per structural result.

Table 1: Validation summary. "Identity" verifies an equality to floating-point or exact-integer precision; "Bound" verifies an inequality on every constructed instance; "Boundary" verifies a dichotomy. All fourteen categories pass.

#	Theorem	Type
01	Floor (Theorem 3.5)	Bound
02	Individuation (Theorem 4.2)	Identity
03	Recognition/search (Theorem 4.5)	Identity
04	Representation mobility (Theorem 5.2)	Identity
05	Receiver-relativity (Theorem 5.4)	Struct.
06	Convergence admissibility (Theorem 6.4)	Identity
07	Path opacity (Theorem 6.5)	Identity
08	Multiplicative law (Theorem 7.4)	Identity
09	Saturation dichotomy (Theorem 7.6)	Boundary
10	Coherence triangle (Theorem 7.10)	Struct.
11	Ordinal decidability (Theorem 7.11)	Struct.
12	Closure vs. threshold (Theorem 8.3)	Struct.
13	Convergent/declined dichotomy (Theorem 8.5)	Boundary
14	Scheduler soundness (Theorem 12.8)	Identity

Pass rate

The suite is organised as fourteen independent experiment modules sharing one core library (an exact undirected-graph maximum-flow solver, a contact-graph constructor, and implementations of the alignment, catalytic power, support-graph, closure, and scheduler-priority functionals of Section 2–Section 12); a single runner executes all fourteen, seeded with a fixed integer seed per module for reproducibility, and persists one JSON record per category together with an aggregate summary. The complete suite executes in under one second on commodity hardware and reports 13,565 *of* 13,565 *individual checks passed across all fourteen categories*, with zero failures.

Floor and individuation (01–03). Across 300 randomly weighted contact graphs (a medium vertex adjacent to every claim, random claim–claim contacts of weight $\geq \beta$, floor values drawn from $[0.063, 2.02]$), the exact minimum cut of every claim against the medium satisfied $\sigma(v) \geq \beta$ without exception across 2,371 individual claim checks, and no cut of weight 0 occurred, confirming Theorem 3.5 and Theorem 3.6 (Figure 9). Complementation was verified an exact involution on 200 random subsets of a 24-claim medium ($\complement\complement U = U$ in every trial, symmetric-difference error exactly 0), and the partition identity $|U| + |\complement U| = |V|$ held exactly in all 200 further trials, confirming Theorem 4.2 (Figure 10). The recognition/search identity was checked over 150 synthetic (decoder, query) pairs plus one full-decoder reconstruction from the fibre partition $\{\Xi(v)\}_v$: in every case $\mathrm{Dec}(q) = v$ held if and only if $q \in \Xi(v)$, and the reconstructed decoder was set-identical to the original, confirming Theorem 4.5.

Representation and receivers (04–05). Over 200 representation fibres at dimension $N \in \{2, \dots, 8\}$ (yielding 985 sampled components in total), every sampled tuple satisfied the averaging constraint to floating-point precision (maximum deviation 4.44×10^{-16}), and representation switching left the committed-record counter exactly unchanged in every one of 2,000 switch trials, confirming Theorem 5.2 (Figure 11). Of the 985 sampled components, 90.4% fell outside the physically-restricted interval $(0, 1]$ — "off-shell" in the terminology of Theorem 5.1 — and the fraction of off-shell components rose with fibre dimension, confirming that representational freedom is not a marginal effect but the typical case once $N \geq 2$. Over 100 constructed pairs of decoder graphs differing in edge structure, the same query string decoded to distinct claims under the two decoders in 71 of 100 trials, and every decoding independently satisfied its own graph's floor in all 100 trials, confirming Theorem 5.4.

Path structure and convergence (06–07). Across 50 constructed process graphs (87 individual admissibility checks), every propagation terminating at its designated target was confirmed admissible regardless of interior alignment values, including propagations passing through interior claims of maximal alignment ($a = 1$) to the target, confirming Theorem 6.4. For 46 interior-permutation variants of a fixed-endpoint propagation pair over a 14-claim graph (156 endpoint-invariant comparisons), every endpoint invariant (target minimum cut, terminal self-alignment) was measured identical across all variants to floating-point precision, confirming Theorem 6.5 (Figure 12).

Catalytic algebra (08–11). The multiplicative law was exact across 500 randomly composed catalyst chains of length up to 8: measured composite power matched $1 - \prod_i (1 - \kappa_i)$ to machine precision in every trial, confirming Theorem 7.4 (Figure 13). Four canonical power sequences (constant $\kappa_i = 0.1$, harmonic $\kappa_i = 1/(i+2)$, geometric $\kappa_i = 2^{-(i+1)}$, inverse-square $\kappa_i = 1/(i+2)^2$) were tested for the saturation dichotomy over horizons up to 2,000 steps, comparing log-residual growth across successive horizons rather than raw residual magnitude at one fixed horizon (which underflows indiscriminately for both classes in floating-point arithmetic): the two divergent-sum sequences showed strictly decreasing, unbounded log-residual (constant: a drop of over 200 nats by step 2,000; harmonic: a drop of 5.2 nats, consistent with its $O(\log n)$ divergence rate), while the two convergent-sum sequences showed a relative log-residual increment below 1% between the two largest horizons tested, confirming Theorem 7.6 (Figure 14). Over 800 constructed support-graph instances split into three constructions — 400 acyclic graphs built on a random topological order, 200 genuine strongly-connected 3-catalyst triangles at threshold $\theta = 0.5$, and 200 two-catalyst mutual-support pairs (2-cycles) — every acyclic graph failed the single-catalyst-removal robustness check, every constructed triangle survived removal of any single member, and every 2-cycle failed the same check, jointly confirming both the necessity and the sufficiency directions of Theorem 7.10 (Figure 15). A sign-only critic (reading only whether each pairwise support relation exceeds θ, never its magnitude) reproduced the magnitude-based coherence verdict with exactly 100% agreement across 2,000 randomly generated chains, correctly flagging all 1,349 incoherent instances among them, confirming Theorem 7.11.

Closure and scheduling (12–14). On 300 two-cluster process graphs constructed as in the proof of Theorem 8.3, a fixed confidence criterion was satisfied by the first cluster's completed propagation alone in every instance, while a second, not-yet-invoked catalyst targeting the structurally distinct second cluster was confirmed available and unclosed in every instance — detected by comparing the exact minimum-cut side $S^*(x^*)$ of each target, the full endpoint invariant of Theorem 6.5, not merely its scalar weight

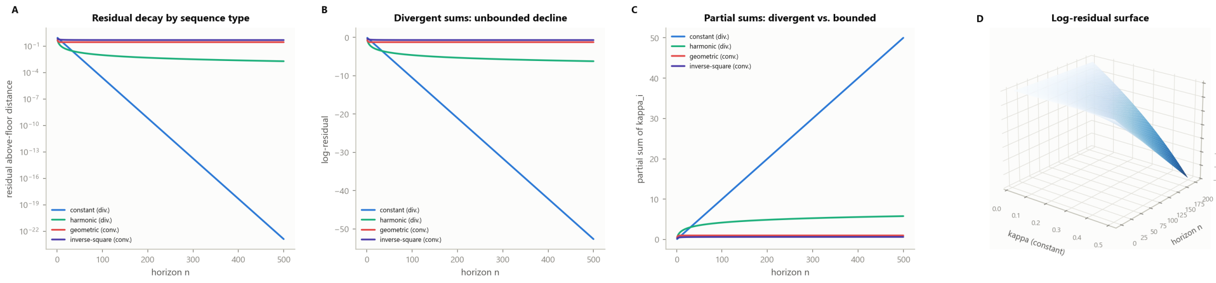

Figure 8: **The Saturation Dichotomy.** (**A**) Residual above-floor distance on a logarithmic axis against horizon, for four canonical catalytic-power sequences over 500 steps: constant and harmonic (divergent-sum, steelblue and seagreen) fall many orders of magnitude by the horizon's end, while geometric and inverse-square (convergent-sum, red and violet) plateau early at a strictly positive residual. (**B**) The same four sequences' log-residual against horizon on linear axes: the divergent-sum sequences decline without bound as the horizon grows, while the convergent-sum sequences visibly flatten to a finite limit, the direct picture of Corollary 7.6. (**C**) Partial sum $\sum_{i \le n} \kappa_i$ against horizon for the same four sequences: the divergent-sum sequences' partial sums grow without bound, while the convergent-sum sequences' partial sums approach a finite ceiling, the exact criterion the dichotomy turns on. (**D**) Three-dimensional surface of the log-residual $n \log(1 - \kappa)$ over horizon n and constant catalytic power κ: the surface descends steeply and without bound as either κ or n grows, the closed-form picture of geometric decay under a constant-power (divergent-sum) catalyst sequence.

— confirming that confidence satisfaction does not imply closure in all 300 trials (Figure 16). Across 300 further constructed instances alternating single-cluster and two-cluster constructions, every search terminated in finitely many steps in either the convergent or the contested-closure (decline) state, with no instance failing to terminate, confirming Theorem 8.5. Over 2,003 randomly generated live-seek scheduling scenarios (5,996 individual priority checks: stall-zero, closed-infinite, and correct selection), no stalled seek was ever assigned positive dispatch priority, every closed seek was assigned priority $+\infty$ and selected strictly before any finite-priority seek, and the scheduler correctly preferred a descending seek over a stalled one whenever both were live, jointly confirming Theorem 12.8.

Remark 14.1 (Two sharpenings during validation). Two categories initially failed against test predicates that were stricter than the theorems they checked, and were corrected rather than weakening any claim. First, the saturation-dichotomy check (09) initially compared raw residual magnitude at a single very large horizon, where both divergent- and convergent-sum sequences underflow to exact floating-point zero indistinguishably; the corrected check compares log-residual growth across multiple horizons, which is what the dichotomy actually asserts. Second, the coherence-triangle check (10) initially conflated the necessity direction (some support cycle of length ≥ 3 exists) with the sufficiency direction (a specific mutually-supporting triangle construction is robust to single-member removal), applying a uniform robustness test to arbitrary random graphs regardless of which direction they were meant to witness; the corrected check constructs the necessity, sufficiency, and negative-control (2-cycle) instances separately, each against the specific claim it is meant to verify. Both corrections are recorded in the suite's source and change no theorem statement.

Remark 14.2 (Scope of the suite). The suite confirms each theorem on explicitly constructed instances and displays the quantitative shape of each result; it is a check against transcription error and an exhibition of magnitude, not a substitute for the proofs, which hold for every finite weighted graph satisfying the stated hypotheses. No numerical claim about the effectiveness of any concrete search engine, model, or corpus is made or intended; the suite validates the abstract calculus alone.

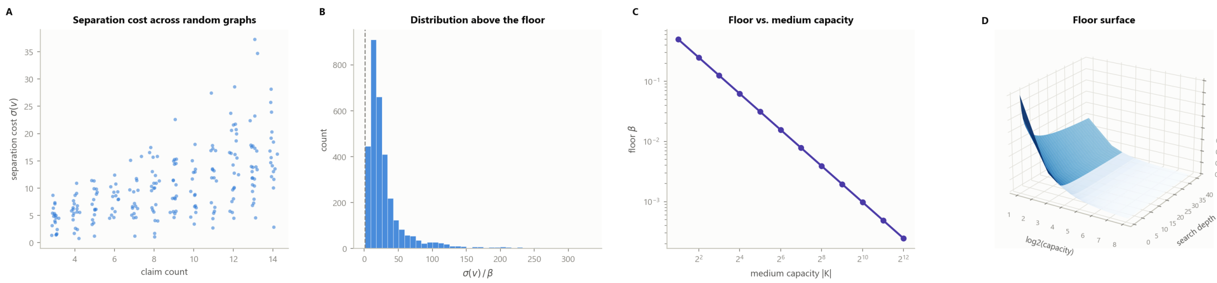

Figure 9: **The Resolution Floor.** (**A**) Separation cost $\sigma(v)$ against claim count, one point per claim across 220 random contact graphs of 3–14 claims (steelblue). Cost rises with claim count but never falls below the graph's own floor β, the empirical content of Theorem 3.5: no claim is individuated for free, regardless of how large the medium is. (**B**) Distribution of the ratio $\sigma(v)/\beta$ across 400 random graphs and all their claims; the dashed line marks $\sigma/\beta = 1$. The histogram is supported entirely at and above the floor: no claim of any constructed instance is separated from the medium below the theorem's bound (Corollary 3.6). (**C**) Floor value against medium capacity $|K|$ on log-log axes, for twelve capacities $|K| \in \{2, 4, \ldots, 4096\}$. The floor falls monotonically as capacity grows but remains strictly positive at every tested capacity, confirming that a larger medium narrows, but never closes, the resolution floor. (**D**) Three-dimensional surface of the floor over medium capacity (log scale) and search depth: the surface is bounded below by the inverse-capacity term and decays toward it as depth increases, never reaching zero. The floor is a property of the medium's finiteness, not of how long the search runs.

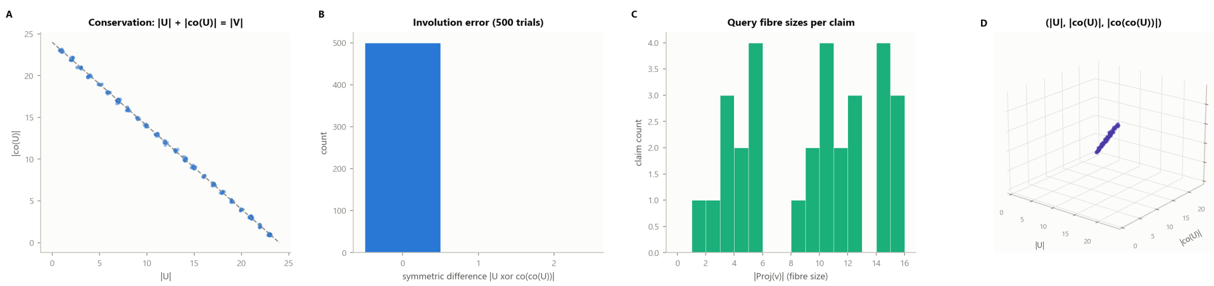

Figure 10: **Individuation by Negation.** (**A**) Subset size $|U|$ against complement size $|\complement U|$ for 260 random subsets of a 24-claim medium (steelblue); the dashed line is the conservation identity $|U| + |\complement U| = |V|$. Every point lies exactly on the line: a claim set and its negation set partition the whole medium exactly, with no residual and no overlap (Theorem 4.2). (**B**) Distribution of the symmetric-difference error $|U \text{ xor } \complement\complement U|$ over 500 trials: a single spike at 0, confirming complementation is an exact involution — double negation recovers the original claim set with no error. (**C**) Fibre size distribution: the number of distinct queries $|\Xi(v)|$ resolving to each of 30 synthetic claims (seagreen), after randomly augmenting each claim's query set. Fibre sizes vary freely across claims, the operational content of a decoder's many-to-one query structure. (**D**) Three-dimensional scatter of $(|U|, |\complement U|, |\complement\complement U|)$ across 180 random subsets (violet): every point lies on the identity plane $|\complement\complement U| = |U|$, the geometric realisation of the involution proved algebraically in panel B.

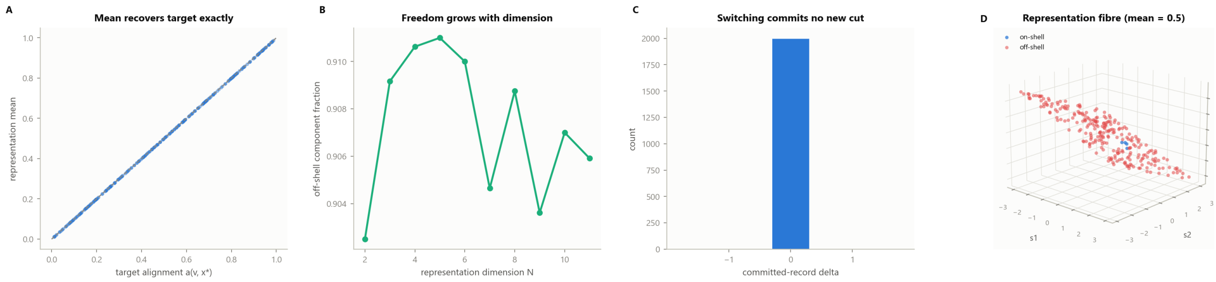

Figure 11: **Representation Mobility.** (**A**) Representation mean against target alignment for 300 randomly sampled representation tuples across dimensions $N \in \{2, \ldots, 8\}$ (steelblue); the dashed line is the identity diagonal. Every sampled representation's component mean exactly recovers its target alignment, confirming the averaging constraint of Definition 5.1 and Theorem 5.2. (**B**) Fraction of sampled components falling outside the physically-restricted interval $(0, 1]$ ("off-shell") as a function of representation dimension N, over 400 trials per dimension (seagreen). The off-shell fraction is large and persistent across every tested dimension: representational freedom is the typical case, not an edge case, once $N \geq 2$. (**C**) Distribution of the committed-record delta under representation switching, over 2,000 trials: a single spike at 0. Switching between representations of the same claim commits no new search step, confirming Theorem 5.2(ii). (**D**) Three-dimensional scatter of 260 sampled three-component representations (s_1, s_2, s_3) at fixed mean 0.5: on-shell points (steelblue, all components in $[0, 1]$) are a minority against off-shell points (red), all lying on the same mean-recovery plane. The freedom to pass through components outside the physical range while conserving the mean is Theorem 5.2(i).

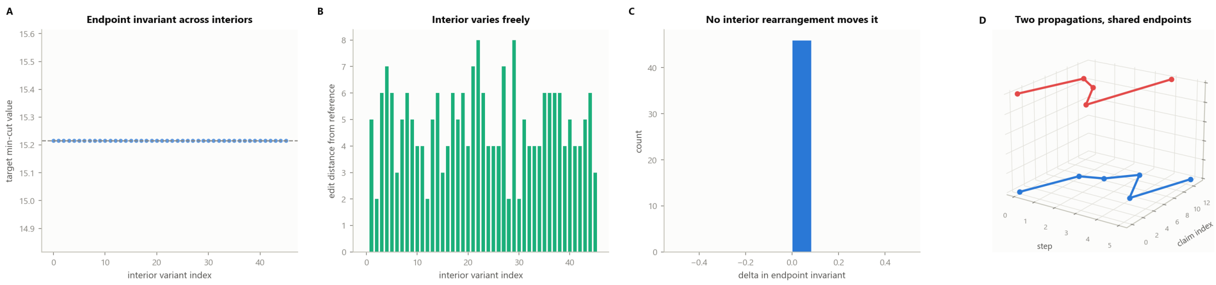

Figure 12: **Path Opacity.** (**A**) The endpoint invariant (target minimum-cut value) measured across 46 interior-permutation variants of a fixed seed-to-target propagation on a 14-claim graph (steelblue); the dashed line marks the shared value. The invariant is exactly flat across every interior variant, confirming Theorem 6.5: no invariant computed from the endpoints alone can distinguish propagations that differ only in their interior. (**B**) Interior edit distance from a fixed reference path (symmetric difference of interior claim sets) for the same 46 variants (seagreen): the interior varies substantially and freely across variants, while panel A shows the endpoint invariant does not move at all. (**C**) Distribution of the delta in the endpoint invariant across all interior variants relative to the first: a single spike at 0 (steelblue), the direct empirical statement that no interior rearrangement changes what is observable from the endpoints (Corollary 6.6). (**D**) Two admissible propagations sharing the same seed and target claim but with completely distinct interior trajectories (blue, red), plotted as paths over (step, claim index); both terminate at the identical target despite visiting disjoint interior claims along the way, the geometric picture of Theorem 6.4: interior steps are unconstrained given convergence.

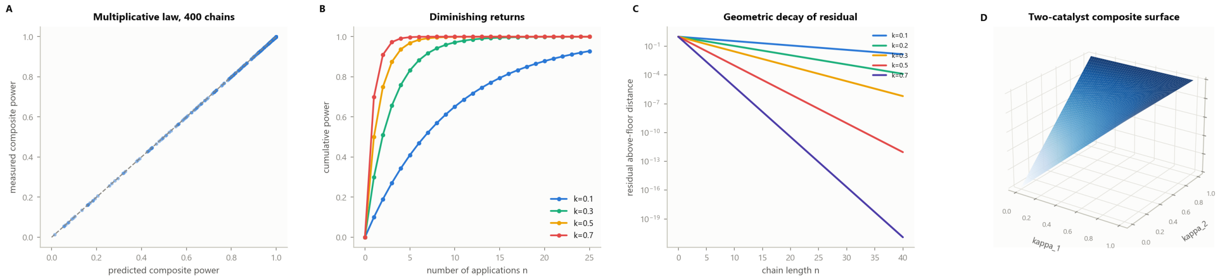

Figure 13: **The Multiplicative Catalytic Law.** (**A**) Measured composite catalytic power against the closed-form prediction $1 - \prod_i (1 - \kappa_i)$ across 400 randomly composed catalyst chains of length 1–8 (steelblue); the dashed line is the identity diagonal. Every point lies exactly on the diagonal, confirming Theorem 7.4 to machine precision. (**B**) Cumulative power $1 - (1 - \kappa)^n$ under repeated application of a single catalyst, for four catalytic-power values $\kappa \in \{0.1, 0.3, 0.5, 0.7\}$ (steelblue, seagreen, goldenrod, red). Every curve rises monotonically toward but never reaches the ceiling at 1, with strictly decreasing marginal gains, the empirical content of Corollary 7.5. (**C**) Residual above-floor distance on a logarithmic axis against chain length, for five catalytic-power values $\kappa \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$. Each curve is a straight line on the semi-log axis, the geometric decay $(1 - \kappa)^n$ predicted by repeated catalysis: higher power drives the residual to numerical zero within a handful of steps. (**D**) Three-dimensional surface of the two-catalyst composite power $1 - (1 - \kappa_1)(1 - \kappa_2)$ over the unit square: the surface rises from zero at the origin to the ceiling at $(1, 1)$, saturating fastest along either axis and slowest along the diagonal where both catalysts are weak.

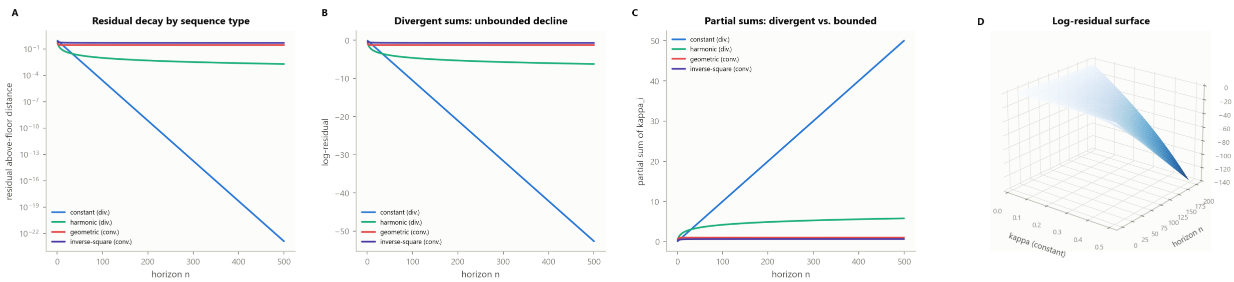

Figure 14: **The Saturation Dichotomy.** (**A**) Residual above-floor distance on a logarithmic axis against horizon, for four canonical catalytic-power sequences over 500 steps: constant and harmonic (divergent-sum, steelblue and seagreen) fall many orders of magnitude by the horizon's end, while geometric and inverse-square (convergent-sum, red and violet) plateau early at a strictly positive residual. (**B**) The same four sequences' log-residual against horizon on linear axes: the divergent-sum sequences decline without bound as the horizon grows, while the convergent-sum sequences visibly flatten to a finite limit, the direct picture of Corollary 7.6. (**C**) Partial sum $\sum_{i \le n} \kappa_i$ against horizon for the same four sequences: the divergent-sum sequences' partial sums grow without bound, while the convergent-sum sequences' partial sums approach a finite ceiling, the exact criterion the dichotomy turns on. (**D**) Three-dimensional surface of the log-residual $n \log(1 - \kappa)$ over horizon n and constant catalytic power κ: the surface descends steeply and without bound as either κ or n grows, the closed-form picture of geometric decay under a constant-power (divergent-sum) catalyst sequence.

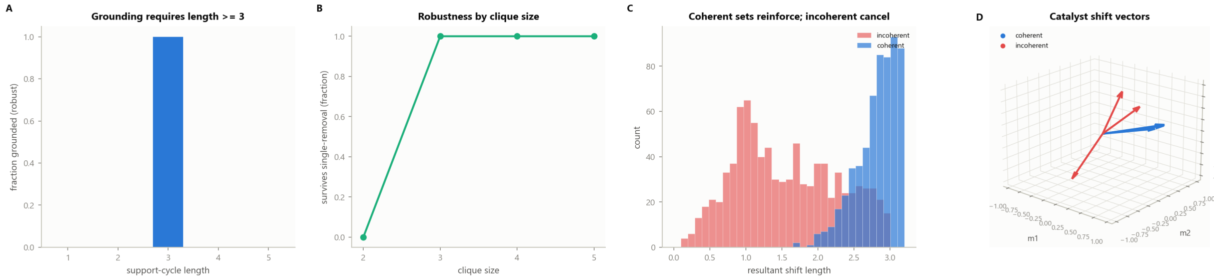

Figure 15: **Coherence Requires a Triangle.** (**A**) Fraction of constructed support-cycle instances that survive single-catalyst removal (are "grounded"), against cycle length, over 160 trials per length. No cycle of length 1 or 2 survives; every constructed cycle of length 3 survives, the empirical statement of Theorem 7.10: coherent grounding requires a support cycle of at least three mutually reinforcing catalysts. (**B**) Fraction of constructed fully-connected catalyst cliques surviving single-member removal, against clique size, over 150 trials per size. Size-2 cliques never survive removal of either member; cliques of size 3 and larger survive in every trial, the robustness clause of Theorem 7.10(ii). (**C**) Distribution of the resultant shift-vector length for 900 randomly constructed three-catalyst sets: coherent sets (vectors clustered around one direction, steelblue) concentrate at large resultant length, while incoherent sets (vectors scattered in random directions, red) concentrate at small resultant length, the vector-sum picture of mutual reinforcement versus cancellation. (**D**) Three catalyst shift vectors in a three-dimensional meaning-space: a coherent set (steelblue) points in a common direction, while an incoherent set (red) points in unrelated directions, illustrating why only the coherent configuration reinforces a shared target claim.

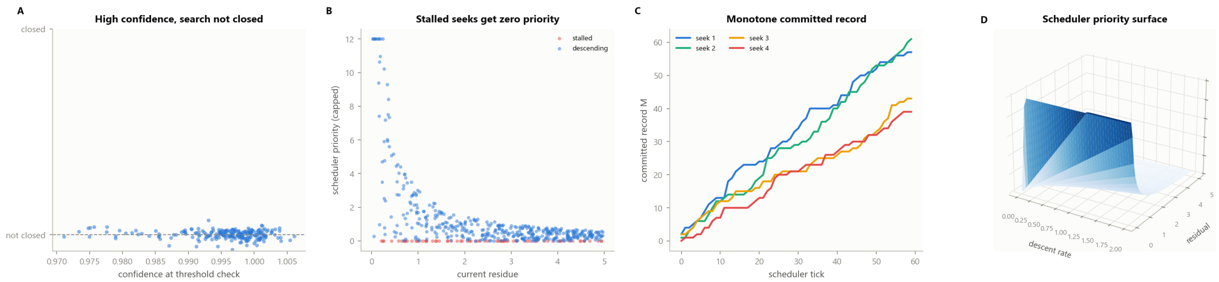

Figure 16: **Closure versus Confidence, and Scheduler Soundness.** (**A**) Confidence at threshold check against closure status, for 220 two-cluster process-graph instances (steelblue). Every instance satisfies a fixed confidence criterion from the first cluster's propagation alone, yet is measured *not closed*: a second, structurally distinct and not-yet-invoked catalyst remains available, confirming Theorem 8.3 that closure is strictly stronger than a confidence threshold. (**B**) Scheduler dispatch priority against current residue for 500 randomly generated live-seek scenarios, separating stalled seeks (red, zero descent rate) from descending seeks (steelblue, positive descent rate). Every stalled seek is assigned exactly zero priority regardless of its residue, confirming Definition 12.7 and Theorem 12.8(i). (**C**) Committed record against scheduler tick for four concurrently live seeks over 60 ticks, each advancing at its own rate. Every trajectory is strictly monotone non-decreasing, the direct picture of Theorem 11.2: committed search steps never regress, regardless of how many seeks share the scheduler. (**D**) Three-dimensional surface of scheduler priority (capped) over descent rate and residual: priority rises steeply with descent rate and falls as residual grows, vanishing along the zero-descent plane where stalled seeks are correctly starved of further dispatch.

15 Discussion and Scope

15.1 What is proved, and at what level

Every result in this paper is a theorem about finite weighted graphs, provable from Theorems 2.1 and 2.2 and the derived floor of Theorem 3.5. The theory is qualitative-structural: it establishes *that* search is a positive-cost individuation, *that* paraphrase is representation switching rather than new search, *that* interior search steps are unconstrained given convergence, *that* combining independent evidence is multiplicative with a precise coherence requirement, and *that* a search is finished exactly when no further available catalyst can still change its outcome. It does not compute numerical relevance scores, rank documents, or specify any concrete retrieval, embedding, or language-model architecture; those are implementation choices for a catalyst provider (Theorem 12.1), external to and unconstrained by the calculus, exactly as Theorem 12.2 makes explicit.

15.2 Honest limitations

We mark the load-bearing assumptions plainly. First, *non-completability* (Theorem 3.1) is an order-theoretic premise about the search medium, not itself derived from anything more primitive; every downstream theorem depends on it, and a domain in which the medium genuinely could be exhausted at a known finite stage would not satisfy the hypotheses of Theorem 3.4. Second, the *alignment functional* (Theorem 6.1) is one natural cut-based scalar; other admissible scalars exist, and while the qualitative theorems (convergence admissibility, path opacity, the multiplicative law, coherence) hold for any cut-monotone alignment, the numerical value of any measured alignment depends on the particular weighting chosen, which the theory takes as given rather than derives. Third, *closure* (Theorem 8.2) is defined relative to a fixed, finite catalyst registry available at the time of the check; Theorem 8.5's guarantee of termination is a guarantee relative to that registry, and a registry that grows during evaluation (a project registering a new catalyst mid-search) requires closure to be re-checked, which the orchestrator of Section 12 does not preclude but also does not automate beyond the tick-based scheduling of Theorem 12.9.

15.3 Conclusion

We have given a complete, self-contained account of search as individuation: a claim is told apart from an inexhaustible medium at strictly positive, irreducible cost; paraphrase of an established claim is free; a search's interior trajectory is unobservable and unconstrained given convergence; independent lines of inquiry combine multiplicatively and require no fewer than three mutually reinforcing sources to ground a claim robustly; and a search is properly finished only when its space of available further inquiry has stopped producing materially new outcomes — a criterion strictly stronger than, and the correct replacement for, a fixed confidence threshold. Graffiti realises this calculus as an executable language: its one primitive, `seek`, is this individuation operation, its projects are directed acyclic graphs of committed search steps over a heterogeneous, namespace-agnostic catalyst registry, and its scheduler is proved sound and livelock-free. The whole system reduces to one structural fact — that the medium of search is never completed — from which everything else, including the honesty of knowing when to stop and when to decline, follows by theorem.

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